# Module 4
# Lecture Notes

Competencies 97 to 132. One section per competency, each headed with its number and what you should be able to do, with worked exam questions at the end of every topic.

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Each topic and every competency inside it. Use the competency number to find the section, and the competency packet to see the exact wording you are held to.


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# The respiratory system

The written notes for the respiratory lectures. Module 4 competencies 97 to 104, with eight worked questions at the end.

## Before the competencies: what the system does

Respiration is four linked jobs. Two of them belong to this system, one belongs to the cardiovascular system, and the last one happens out in the tissues. The respiratory system and the cardiovascular system are so tightly linked that if either fails, oxygen never reaches the cells.

1. Pulmonary ventilation (breathing): air moves into the lungs (inspiration) and out of the lungs (expiration), like a bellows.
2. External respiration: oxygen crosses from the alveoli into the blood and carbon dioxide crosses from the blood into the alveoli, at the respiratory membrane.
3. Transport of gases: the cardiovascular system carries oxygen from the lungs to the tissues and carbon dioxide back. The heart and vessels are the delivery truck; the blood is the cargo.
4. Internal respiration: oxygen moves from the systemic capillaries into the tissue cells and carbon dioxide moves out of them.

- Cellular respiration is something else: the mitochondria using oxygen and making carbon dioxide to produce ATP. It is a job of every cell, not of the respiratory system.
- Moving air also gives you smell and speech.
- The structures in order, nose to alveolus: nose, nasal cavity, paranasal sinuses, pharynx, larynx, trachea, bronchi and their branches, lungs, alveoli. (Nice Nurses Practice Thorough Bronchial Lung Assessments.)
- The respiratory muscles, the diaphragm and its helpers, create the pressure changes that move the air. They are competency 103.

## Competency 97. Upper and lower tracts, conducting and respiratory zones

**You should be able to:** Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones, and assign the major airway structures to each.

The respiratory system moves air into and out of the body and brings that air close to the blood. Air follows one continuous path from the nose to the air sacs of the lungs. Anatomists group that path two ways, and the two groupings use different dividing lines, so you have to learn both.

### One breath, step by step

1. Air enters at the nares (nostrils) and passes the nasal vestibule, where coarse hairs catch large particles.
2. It moves through the nasal cavity, tumbling past the conchae, and is warmed and moistened.
3. It leaves through the choanae into the nasopharynx (air only), then the oropharynx and laryngopharynx (shared with food).
4. It turns forward into the larynx and passes the glottis, between the vocal folds.
5. It runs down the trachea to the carina, where the trachea forks.
6. It enters a main bronchus (one per lung), then a lobar bronchus (one per lobe), then a segmental bronchus (one per bronchopulmonary segment).
7. It passes through bronchioles, which have no cartilage, to a terminal bronchiole, the last part of the conducting zone.
8. It crosses into a respiratory bronchiole, the first airway of the respiratory zone, then an alveolar duct, an alveolar sac, and an alveolus, where gases cross into the blood.

**Upper respiratory tract**
: The nose, nasal cavity, paranasal sinuses, and pharynx. These sit in the head and neck.

**Lower respiratory tract**
: The larynx, trachea, bronchi and all their branches, and the lungs. The larynx is the first structure of the lower tract, so the line between the tracts falls where the pharynx meets the larynx. Some clinical sources put the larynx in the upper tract; these notes follow the anatomy textbook split.

**Conducting zone**
: Every passage that only carries air, from the nose down to the terminal bronchioles. Its walls are too thick for gases to cross. It warms, moistens, and filters the air on the way through.

**Respiratory zone**
: Where gas exchange happens: the respiratory bronchioles, alveolar ducts, alveolar sacs, and alveoli. The line between the zones is deep inside the lung, where a terminal bronchiole gives way to a respiratory bronchiole.

| Structure | Tract | Zone |
|---|---|---|
| **Nose and nasal cavity** | Upper | Conducting |
| **Paranasal sinuses** | Upper | Air spaces off the nasal cavity, no gas exchange |
| **Pharynx** | Upper | Conducting |
| **Larynx** | Lower | Conducting |
| **Trachea** | Lower | Conducting |
| **Main, lobar, and segmental bronchi** | Lower | Conducting |
| **Bronchioles and terminal bronchioles** | Lower | Conducting |
| **Respiratory bronchioles** | Lower | Respiratory |
| **Alveolar ducts and sacs** | Lower | Respiratory |
| **Alveoli** | Lower | Respiratory |

- Tract tells you where a structure is. Zone tells you what it does.
- Because the lines fall in different places, one structure can be lower tract and conducting zone at the same time. The trachea is the usual example.
- Every upper tract structure is conducting. The lower tract holds the rest of the conducting zone plus the whole respiratory zone.
- The conducting zone ends at the terminal bronchioles, not at the bronchi and not where the lung begins. Gas exchange itself is physiology; this course covers the structures.

## Competency 98. Nose, nasal cavity, and pharynx

**You should be able to:** Identify the parts of the nasal cavity, paranasal sinuses, and the three regions of the pharynx.

The nose is the main entrance for air. It warms, moistens, and filters the air, it houses the receptors for smell, and it acts as a resonating chamber for the voice. Bone forms the upper part of the external nose (nasal bones, frontal bone, maxillae) and hyaline cartilage forms the lower part, which is why the tip bends.

### The external nose

| Feature | What it is |
|---|---|
| **Root** | Where the nose meets the forehead, between the eyebrows (the glabella is the flat area between the brows) |
| **Bridge** | The upper, bony part |
| **Dorsum nasi** | The anterior margin, from the root to the apex |
| **Apex** | The tip |
| **Alae** | The flared sides of the nostrils |
| **Nares** | The nostrils, the external openings |

- Framework: nasal bones and the frontal bone above, the maxillae at the sides, and plates of hyaline cartilage below (septal and alar cartilages and the lateral processes of the septal cartilage).
- The nose does five things: it is an airway, it warms and moistens the air, it filters and cleans it, it resonates for speech, and it houses the smell receptors.

### Parts of the nasal cavity

**Nasal septum**
The midline wall that splits the cavity into right and left halves. Septal cartilage in front; the perpendicular plate of the ethmoid above the vomer behind.

**Nasal vestibule**
The area just inside each nostril, lined with skin, sebaceous glands, and coarse hairs that trap large particles.

**Nasal conchae**
Three scroll-shaped shelves on each lateral wall: superior, middle, and inferior. The superior and middle conchae are parts of the ethmoid bone. The inferior concha is a separate bone. They make the air tumble so more of it touches the warm, moist lining.

**Nasal meatuses**
The passages under each concha: superior, middle, and inferior meatus. A meatus is the space, not the shelf.

**Hard palate**
The bony front part of the floor, separating the nasal cavity from the mouth.

**Soft palate**
The muscular back part of the floor. It lifts to close off the nasopharynx during swallowing.

**Choanae**
The two back openings where the nasal cavity becomes the nasopharynx.

### Two linings

| Mucosa | Epithelium | Job |
|---|---|---|
| **Olfactory mucosa** | Olfactory epithelium, a small patch in the roof | Houses the smell receptors |
| **Respiratory mucosa** | Pseudostratified ciliated columnar with goblet cells | Secretes mucus that traps dust and microbes; the cilia sweep it back to the pharynx to be swallowed |

- The vestibule is lined with stratified squamous epithelium, because it takes the mechanical wear at the entrance.
- The seromucous glands secrete lysozyme, an antibacterial enzyme, and the epithelial cells secrete defensins. The high water content of the mucus humidifies the air.
- A dense capillary bed close to the surface warms incoming air. It is also why nosebleeds (epistaxis) are common.
- Nerve endings in the lining trigger the sneeze reflex when dust or pollen irritates them.
- Cold air slows the cilia, so mucus stops moving back toward the throat and drips out the front. That is why your nose runs in cold weather. On the way out, water vapor condenses on the cooler lining, so the conchae reclaim heat and moisture during exhalation.

### The paranasal sinuses

Air-filled spaces inside four skull bones around the nasal cavity, each named for its bone and lined with the same mucosa as the nasal cavity. Each drains through a small opening into the nasal cavity. They lighten the skull, help warm and moisten air, and add resonance to the voice.

| Sinus | Bone | Where it sits | Drains into |
|---|---|---|---|
| **Frontal sinus** | Frontal | In the forehead, above the orbits | Middle meatus |
| **Ethmoidal air cells** | Ethmoid | Many small spaces between the nasal cavity and the orbits | Middle meatus (front and middle cells), superior meatus (back cells) |
| **Sphenoidal sinus** | Sphenoid | Deep in the skull, behind the nasal cavity | The sphenoethmoidal recess, above the superior concha |
| **Maxillary sinus** | Maxilla | In the cheek, below each orbit; the largest | Middle meatus |

### The pharynx, top to bottom

The pharynx (throat) is a funnel-shaped muscular tube from the base of the skull down to the level of the cricoid cartilage, where it continues as the esophagus. It connects the nasal cavity and mouth to the larynx and esophagus.

| Region | Location | Carries | Epithelium and structures |
|---|---|---|---|
| **Nasopharynx** | Behind the nasal cavity, above the soft palate | Air only | Pseudostratified ciliated columnar. Pharyngeal tonsil (adenoid) on the back wall, auditory tube openings on the side walls |
| **Oropharynx** | Behind the oral cavity, soft palate to epiglottis | Air and food | Stratified squamous, which resists friction. Palatine tonsils in the side walls, lingual tonsil on the back of the tongue |
| **Laryngopharynx** | Behind the larynx, epiglottis to esophagus | Air and food | Stratified squamous. Where air turns forward into the larynx and food goes back into the esophagus |

- When you swallow, the soft palate and uvula lift and seal the nasopharynx, so food does not enter the nose.
- The epithelium changes with the job. Where only air passes, the lining is ciliated and moves mucus. Where food also passes, it is stratified squamous and stands up to rubbing.

## Competency 99. Larynx

**You should be able to:** Identify the cartilages and folds of the larynx, including the thyroid and cricoid cartilages, epiglottis, arytenoids, vocal cords, and glottis.

The larynx (voice box) connects the laryngopharynx to the trachea. It sits in the front of the neck at about C3 to C6 and hangs from the hyoid bone, which is above it and not part of it. It has three jobs: hold the airway open, keep food and drink out of the lower airway, and produce the voice.

### The nine cartilages

| Cartilage | Number | Cartilage type | What it is |
|---|---|---|---|
| **Thyroid cartilage** | Single | Hyaline | The large shield. Its front ridge is the laryngeal prominence (Adam's apple). Open at the back |
| **Cricoid cartilage** | Single | Hyaline | The signet ring just below the thyroid cartilage, narrow in front and wide behind. The only complete ring of cartilage in the airway. The trachea hangs from its lower edge |
| **Epiglottis** | Single | Elastic | The spoon-shaped flap attached inside the thyroid cartilage. It tips down over the laryngeal inlet during a swallow |
| **Arytenoid cartilages** | Paired | Mostly hyaline | Sit on top of the back of the cricoid. They anchor the vocal folds and pivot to move them, so they are the most important paired cartilages |
| **Corniculate cartilages** | Paired | Elastic | Small cartilages on top of the arytenoids |
| **Cuneiform cartilages** | Paired | Elastic | Small cartilages in the fold running from each arytenoid up to the epiglottis |

- The cricothyroid ligament fills the gap between the thyroid and cricoid cartilages in front. It lies just under the skin, below the vocal folds, with no cartilage over it, which is why it is the landmark for an emergency airway when the larynx above is blocked.
- The epiglottis is elastic because it bends over the inlet with every swallow and springs back. The thyroid, cricoid, and arytenoid cartilages are hyaline (only the tips of the arytenoids are elastic).

### The folds and the glottis

**Vocal folds**

The true vocal cords, the lower pair. Each contains a vocal ligament, an elastic band running from an arytenoid cartilage behind to the thyroid cartilage in front. Very few blood vessels, so they look pearly white. They vibrate as air passes between them, and that vibration is sound.

**Vestibular folds**

The false vocal cords, the upper pair. They help close the airway during swallowing and play no part in sound.

**Glottis**

The vocal folds together with the opening between them (the rima glottidis), the narrowest point air passes in the larynx. Books draw the edges a little differently: some use the name for the opening alone, and OpenStax includes the vestibular folds.

**Laryngeal inlet**

The opening into the larynx from the laryngopharynx, which the epiglottis covers during a swallow.

- When the arytenoids pivot, the vocal folds swing apart to widen the glottis for breathing or come together to close it.
- The lining changes at the vocal folds. Above them, where food can rub, stratified squamous. Below them, pseudostratified ciliated columnar, with cilia sweeping mucus up toward the pharynx.
- The vocal folds make the sound, but the pharynx, oral cavity, and sinuses amplify it like the body of an instrument, and the lips, tongue, and soft palate shape it into words.
- The larynx also works as a sphincter. When you strain, lift something heavy, or empty your bowels, the vocal folds close and hold air in the chest (the Valsalva maneuver), which stiffens the trunk and raises abdominal pressure.
- The laryngeal prominence grows larger in males at puberty under testosterone, which is why the voice drops.

## Competency 100. Trachea and bronchial tree

**You should be able to:** Trace air through the trachea and bronchial tree in order from the trachea and carina down to the terminal bronchioles.

Below the larynx the airway becomes a branching tree. The tubes get narrower and more numerous at every step, and every one of them down to the terminal bronchioles belongs to the conducting zone.

### From the trachea to a terminal bronchiole, in order

1. Trachea: the windpipe, from the lower edge of the cricoid cartilage into the mediastinum, about 10 to 12 cm long. Sixteen to twenty C-shaped rings of hyaline cartilage hold it open. The open side of each C faces back toward the esophagus, and the trachealis muscle bridges the gap, so the esophagus can bulge forward when you swallow.
2. Carina: the internal ridge formed by the lowest tracheal cartilage at the fork, about at the sternal angle (T4 to T5). Its lining is very sensitive; touching it sets off a strong cough.
3. Main (primary) bronchi: one enters each lung at the hilum. The right main bronchus is wider, shorter, and more vertical than the left, so an inhaled object usually lands in the right lung.
4. Lobar (secondary) bronchi: one to each lobe, three on the right and two on the left.
5. Segmental (tertiary) bronchi: one to each bronchopulmonary segment.
6. Bronchioles: tubes less than 1 mm across, with smooth muscle and no cartilage.
7. Terminal bronchioles: the last and smallest conducting tubes. The next tubes, the respiratory bronchioles, begin the respiratory zone.

### The four layers of the tracheal wall

| Layer | What is in it |
|---|---|
| **Mucosa** | Pseudostratified ciliated columnar epithelium with goblet cells on a lamina propria rich in elastic fibers. The cilia move mucus up toward the pharynx |
| **Submucosa** | Connective tissue with seromucous glands that produce sheets of mucus |
| **Cartilaginous layer** | The 16 to 20 C-shaped rings of hyaline cartilage, open at the back, bridged by the trachealis muscle. The last cartilage forms the carina |
| **Adventitia** | The outer connective tissue that ties the trachea to its neighbors |

- The elastic fibers let the trachea stretch and recoil with each breath.
- When the trachealis contracts it narrows the trachea, which speeds the air during a cough and helps blast secretions out.

### How the wall changes down the tree

| Level | Cartilage | Epithelium | Smooth muscle |
|---|---|---|---|
| **Trachea** | C-shaped hyaline rings | Pseudostratified ciliated columnar with goblet cells | Trachealis across the back of each ring |
| **Main bronchi** | C-shaped rings | Pseudostratified ciliated columnar with goblet cells | A thin layer |
| **Lobar and segmental bronchi** | Irregular plates | Pseudostratified ciliated columnar, fewer goblet cells in smaller branches | A more complete layer |
| **Bronchioles** | None | Simple columnar to simple cuboidal | A complete layer, much of the wall |
| **Terminal bronchioles** | None | Simple cuboidal | A complete layer |

- Going down the tree, cartilage shrinks and smooth muscle takes over. Rings, then plates, then none.
- Cartilage is already gone at the bronchioles, not just at the terminal bronchioles.
- Without cartilage to hold them open, bronchioles narrow a great deal when their smooth muscle contracts. That is the structural reason asthma is so dangerous.

## Competency 101. Lungs and pleurae

**You should be able to:** Identify the lobes, fissures, and surfaces of each lung and the parietal and visceral pleurae with the pleural cavity.

The two lungs fill most of the thoracic cavity, one on each side of the mediastinum (the central compartment holding the heart, great vessels, trachea, and esophagus), with the diaphragm beneath them. Each lung is cone-shaped, soft, and spongy. The two are not the same size: the heart tilts left and takes room, so the left lung is smaller; the liver pushes the right dome of the diaphragm up, so the right lung is shorter and wider.

### Surfaces and landmarks

**Apex**
The narrow top of the lung, rising just above the clavicle into the root of the neck.

**Base**
The broad, curved bottom surface resting on the diaphragm.

**Costal surface**
The large outer surface against the ribs.

**Mediastinal surface**
The medial surface facing the mediastinum. The hilum is on it.

**Hilum**
The indented area on the mediastinal surface where the bronchus, blood vessels, nerves, and lymphatic vessels enter and leave.

**Root of the lung**
The bundle passing through the hilum: main bronchus, pulmonary artery, pulmonary veins, bronchial vessels, nerves, and lymphatics.

**Cardiac notch**
The indentation in the front edge of the left lung that makes room for the heart.

**Lingula**
The tongue-shaped part of the left superior lobe just below the cardiac notch. It sits where the middle lobe sits on the right.

- In the root, the pulmonary artery carries oxygen-poor blood into the lung and the pulmonary veins carry oxygen-rich blood back to the heart.
- Telling right from left at the hilum on the cadaver: in the left hilum the pulmonary artery is the highest structure; in the right hilum a bronchus sits at or above the artery. In both, the pulmonary veins are lowest.

### Lobes and fissures

| Lung | Lobes | Fissures | Notes |
|---|---|---|---|
| **Right lung** | Three: superior, middle, inferior | Oblique fissure (separates the inferior lobe) and horizontal fissure (separates superior from middle) | Larger; shorter and wider because the liver raises the right diaphragm. Ten bronchopulmonary segments |
| **Left lung** | Two: superior, inferior | Oblique fissure only | Smaller, with the cardiac notch and the lingula. About 8 to 10 segments |

- Each lobe is divided into bronchopulmonary segments, each served by its own segmental bronchus, artery, and vein. Because each segment is its own unit, a surgeon can remove a diseased segment without damaging the segments around it, like taking one section out of an orange.
- Right has three, left has two. The left gave up its middle lobe to make room for the heart, and the lingula is what is left in its place.

### Two blood supplies to the lung

| Circulation | Pressure and volume | Job |
|---|---|---|
| **Pulmonary circulation (pulmonary arteries and veins)** | Low pressure, high volume | Gas exchange at the alveoli |
| **Bronchial circulation (bronchial arteries off the aorta)** | High pressure, low volume | Oxygen-rich systemic blood to feed the lung tissue itself |

The lungs are supplied by parasympathetic and sympathetic motor fibers and by visceral sensory fibers. The sympathetic fibers widen the airways; the parasympathetic fibers narrow them.

### The pleurae

**Parietal pleura**
The serous membrane lining the inside of the thoracic wall, the top of the diaphragm, and the sides of the mediastinum.

**Visceral pleura**
The serous membrane clinging to the lung surface and dipping into the fissures. The two layers are one continuous sheet that turns back on itself at the hilum.

**Pleural cavity**
The very thin space between the two layers, holding only a film of pleural fluid. There is a right and a left pleural cavity, kept apart by the mediastinum.

**Costodiaphragmatic recess**
The low pocket where the parietal pleura turns from the ribs onto the diaphragm. The lowest part of the pleural cavity, where extra fluid collects.

- The fluid does two jobs: it lets the lung glide against the chest wall, and it holds the two layers together so the lung follows the chest wall as it expands.
- Because the two pleural cavities are separate, air or fluid in one does not collapse the other lung.

## Competency 102. Respiratory zone and alveolar structure

**You should be able to:** Describe the structure of an alveolus and identify the type I and type II alveolar cells, alveolar macrophages, respiratory membrane, and pulmonary capillaries.

Past the terminal bronchioles the airway walls thin into tiny air sacs. The respiratory zone is the bulk of the lung tissue, and at low power it looks lacy because it is mostly air spaces separated by thin walls.

**Respiratory bronchioles**
The first airways of the respiratory zone, with a few alveoli budding from their walls.

**Alveolar ducts**
Passages whose walls are lined entirely with alveoli.

**Alveolar sacs**
Clusters of alveoli opening into a shared space, like a bunch of grapes.

**Alveoli**
The single tiny air sacs where gas exchange happens. The two lungs hold hundreds of millions of them.

**Alveolar pores**
Small openings connecting neighboring alveoli.

### The alveolar wall and the respiratory membrane

| Structure | What it is | Job |
|---|---|---|
| **Type I alveolar cells** | Very thin, flat simple squamous cells covering nearly all of the alveolar surface | Gases cross the wall through them |
| **Type II alveolar cells** | Rounder, cuboidal cells scattered among the type I cells | Secrete surfactant, which lowers surface tension and keeps the alveoli from collapsing; also secrete antimicrobial proteins |
| **Alveolar macrophages** | The dust cells, moving freely over the inner surface of the alveoli | Engulf dust, microbes, and debris that got past the conducting zone |
| **Pulmonary capillaries** | A dense net wrapping each alveolus, fed by branches of the pulmonary arteries and drained by the pulmonary veins | Bring blood within a membrane's thickness of the air |
| **Respiratory membrane** | The barrier between alveolar air and blood where a capillary presses on an alveolus | Three layers, air to blood: the type I alveolar cell, the fused basement membranes of the alveolus and capillary, and the capillary endothelial cell |

- Type I is thin, for exchange. Type II makes surfactant. Macrophages clean up.
- Elastic fibers surround each alveolus and let the lung stretch on the way in and recoil on the way out.
- The membrane is extremely thin and the alveoli are extremely many, so the lungs have a huge surface for gas exchange. That surface is fragile, which is the story of emphysema.

## Competency 103. Thoracic wall and diaphragm

**You should be able to:** Identify the diaphragm, the intercostal muscles, and the three structures passing through the diaphragm at T8, T10, and T12.

The lungs sit inside the thoracic wall: ribs and costal cartilages, the sternum in front, the thoracic vertebrae behind, muscle between the ribs, and the diaphragm closing the bottom. Breathing works by changing the size of this cage. The lungs follow the wall because the pleural fluid holds them to it.

### The diaphragm

- A dome-shaped sheet of skeletal muscle forming the floor of the thoracic cavity and separating it from the abdominal cavity. Its fibers attach around the lower rib cage, the sternum, and the lumbar vertebrae and pull toward the central tendon, a flat sheet at the top of the dome.
- The right dome sits higher because the liver is under it.
- When it contracts it flattens and moves down, making the thoracic cavity taller. It is the main muscle of breathing.
- Supplied by the right and left phrenic nerves, from spinal nerves C3, C4, and C5. On the cadaver each phrenic nerve runs down the side of the pericardium, in front of the root of the lung. C3, 4, and 5 keep the diaphragm alive.

### The intercostal muscles and the accessory muscles

| Muscle | Fiber direction or shape | Role in breathing | Nerve |
|---|---|---|---|
| **Diaphragm** | Dome, fibers rising to the central tendon | Main muscle of inspiration; contracts, flattens, moves down | Phrenic nerves (C3 to C5) |
| **External intercostals** | Downward and forward, like hands sliding into front pockets | Lift the ribs in inspiration, quiet and forced | Intercostal nerves |
| **Internal intercostals** | At right angles to the externals, downward and backward | Pull the ribs down in forced expiration (a hard cough, blowing out candles) | Intercostal nerves |

- Quiet expiration needs no muscle effort; the lungs and chest wall recoil on their own.
- Harder inspiration recruits the scalenes, sternocleidomastoid, and pectoralis minor. Forced expiration uses the internal intercostals and the abdominal muscles.

### Three structures through the diaphragm

| Vertebral level | Opening | Main structure | Also passes through |
|---|---|---|---|
| **T8** | Caval opening, in the central tendon | Inferior vena cava | |
| **T10** | Esophageal hiatus, in the muscle | Esophagus | Vagus nerves |
| **T12** | Aortic hiatus, behind the diaphragm between the crura | Aorta | Thoracic duct |

- I-8, E-10, A-12. Inferior vena cava at T8, Esophagus at T10, Aorta at T12. These are vertebral levels, listed top to bottom. (I Eat At 8, 10, 12 o'clock.)
- Above the diaphragm the vessel is the thoracic aorta; below it, the abdominal aorta. The crura are the muscular legs that anchor the diaphragm to the lumbar vertebrae.

### Vessels you will find in the upper thorax (lab)

| Structure | Artery or vein | Where it is |
|---|---|---|
| **Brachiocephalic trunk** | Artery | First branch of the aortic arch, right side only; splits into the right subclavian and right common carotid |
| **Common carotid arteries** | Artery | Up the neck to the head; the right from the brachiocephalic trunk, the left straight off the arch |
| **Subclavian arteries** | Artery | Out to the upper limbs; the right from the brachiocephalic trunk, the left straight off the arch |
| **Internal jugular veins** | Vein | Drain the head, down the neck beside the common carotids |
| **Subclavian veins** | Vein | Drain the upper limbs |
| **Brachiocephalic veins** | Vein | Each forms where an internal jugular joins a subclavian vein; the two join to form the superior vena cava |
| **Pulmonary arteries** | Artery | From the heart into each lung at the hilum |
| **Pulmonary veins** | Vein | From each lung back to the heart |

For each one, say the side and whether it is an artery or a vein before you name it.

## Competency 104. Respiratory disorders by structure affected

**You should be able to:** Name common respiratory disorders and state which airway, alveolar, or pleural structure each one affects.

Most respiratory disorders do one of three things. Some narrow or block the airways. Some damage or fill the alveoli, so the exchange surface is lost. Some affect the pleurae and pleural cavity, so the lung cannot stay against the chest wall. Learn the structure first, then what happens to it.

| Disorder | Structure affected | What changes in that structure |
|---|---|---|
| **Rhinitis** | Nasal mucosa | The lining is inflamed, swollen, and makes extra mucus (allergies or infection) |
| **Sinusitis** | Paranasal sinuses | The sinus lining is inflamed, often when the openings into the nasal cavity are blocked |
| **Laryngitis** | Larynx, vocal folds | The vocal folds are inflamed and swollen, cannot vibrate normally, and the voice turns hoarse |
| **Asthma** | Bronchi and bronchioles | The lining is inflamed, smooth muscle tightens, extra mucus collects, and the airways narrow. Bronchioles have no cartilage to hold them open |
| **Chronic bronchitis** | Bronchi | The lining stays inflamed and makes too much mucus, which blocks the airways |
| **Cystic fibrosis** | Bronchi and bronchioles | An inherited disorder; the mucus is thick and sticky, the cilia cannot clear it, and it clogs the airways |
| **Emphysema** | Alveoli | Alveolar walls break down and many small alveoli merge into fewer, larger spaces. The exchange surface shrinks and the lung loses elastic recoil. The change is permanent |
| **COPD** | Airways and alveoli | Chronic obstructive pulmonary disease, a long-term blockage of airflow; mainly emphysema and chronic bronchitis together |
| **Pneumonia** | Alveoli | An infection inflames the alveoli and fills them with fluid. The walls are not destroyed |
| **Tuberculosis** | Lung tissue | A bacterial infection that forms small nodules (tubercles) and scar tissue |
| **Infant respiratory distress syndrome** | Alveoli (type II cells) | A premature newborn's type II cells do not yet make enough surfactant, so the alveoli collapse with each breath |
| **Lung cancer** | Lung tissue | A malignant tumor, usually starting in the epithelium lining the airways or alveoli; strongly linked to smoking |
| **Pleurisy** | Pleurae | The pleural layers are inflamed and rough, so they rub with each breath and cause sharp pain |
| **Pleural effusion** | Pleural cavity | Extra fluid collects in the pleural cavity and presses on the lung; in an upright person it pools in the costodiaphragmatic recess |
| **Pneumothorax** | Pleural cavity | Air enters the pleural cavity, the layers separate, and the lung on that side collapses |

### What this means for patients

- Asthma narrows the bronchioles so much because they have no cartilage to hold them open.
- Emphysema and pneumonia both take away working alveoli, but only emphysema destroys the walls. Pneumonia fills the sacs; emphysema erases them.
- A pneumothorax collapses one lung, not both, because each lung has its own sealed pleural cavity. The same thin, sealed space that lets the lung glide is what makes it vulnerable.
- Because the right main bronchus is wider, shorter, and more vertical, an inhaled object usually ends up in the right lung.
- The cricothyroid ligament sits just under the skin below the vocal folds with no cartilage over it, so it is the landmark for an emergency airway when the larynx above is blocked.

# The lymphatic system

The written notes for the lymphatic system lecture: the one-way lymph pathway, the structure of lymphatic vessels and lymph nodes, the primary and secondary lymphoid organs, and the common lymphatic disorders. Module 4 competencies 105 to 108, with eight worked questions.

## Competency 105. Lymph pathway

**You should be able to:** Define lymph and trace its one-way route from lymphatic capillaries through collecting vessels, nodes, trunks, and ducts back to the subclavian veins.

Blood capillaries leak. All day long, fluid filters out of them into the spaces between cells, and that fluid is called interstitial fluid. The blood capillaries take most of it back, but some fluid and a small amount of plasma protein stay behind in the tissues. The lymphatic system collects this leftover fluid, about 3 liters a day, and returns it to the blood. That keeps fluid from building up in the tissues and keeps blood volume steady.

Once interstitial fluid enters a lymphatic vessel, it is called lymph. Lymph is clear and watery. It carries water, plasma proteins, and whatever else it picks up in the tissues, such as cell debris, bacteria, and sometimes cancer cells. Lymph coming from the small intestine after a fatty meal looks milky white. This fat-rich lymph is called chyle.

### The fluids and the system

**Lymphatic system**
A network of vessels, tissues, and organs that drains tissue fluid and houses the body's defenses.

**Interstitial fluid**
The fluid in the spaces between cells, filtered out of the blood capillaries. In the tissue spaces it is still interstitial fluid, not lymph.

**Lymph**
Interstitial fluid once it has entered a lymphatic vessel. It becomes lymph the moment it enters a lymphatic capillary. Clear and watery.

**Chyle**
Milky, fat-rich lymph coming from the lacteals of the small intestine.

**Three functions**
Returns excess interstitial fluid to the blood, absorbs dietary fats from the small intestine, and houses the immune defenses.

### The one-way route, in order

The lymphatic system is a one-way system. It does not form a loop the way blood vessels do. Lymph starts in the tissues and flows only toward the heart. As it travels, small vessels join into larger and larger vessels, the way small streams join to form a river. On the way, the lymph passes through lymph nodes, which filter it before it reaches the blood.

1. Lymphatic capillaries: blind-ended tubes in the tissues take up excess interstitial fluid. The fluid is now lymph.
2. Lymphatic collecting vessels: larger vessels with valves carry the lymph onward and keep it moving one way.
3. Lymph nodes: nodes spaced along the collecting vessels filter the lymph. Lymph usually passes through several nodes in a row.
4. Lymphatic trunks: collecting vessels merge into large trunks. Each trunk drains one major region of the body.
5. Lymphatic ducts: the trunks empty into one of two ducts, the right lymphatic duct or the larger thoracic duct.
6. Subclavian veins: each duct empties into the blood where the subclavian vein meets the internal jugular vein on its own side. At this point lymph becomes part of the blood plasma again.

- Worked example from the note sheet, starting at the left foot: a lymphatic capillary in the foot, a collecting vessel with valves running up the leg, an inguinal node in the groin, the left lumbar trunk running up the back wall of the abdomen, the cisterna chyli, the thoracic duct up through the thorax, and finally the junction of the left subclavian and left internal jugular veins. Every arrow points toward the heart and none point back.

### The nine trunks

There are nine lymphatic trunks. Each one is named for the region it drains. Four of them come in left and right pairs, and one, the intestinal trunk, is single.

| Trunk | Region drained | Empties into |
|---|---|---|
| **Lumbar (left and right)** | Lower limbs, pelvis, and abdominal wall | Thoracic duct, at the cisterna chyli |
| **Intestinal (single)** | Digestive organs of the abdomen | Thoracic duct, at the cisterna chyli |
| **Bronchomediastinal (left and right)** | Thoracic organs and the thoracic wall | Left: thoracic duct. Right: right lymphatic duct |
| **Subclavian (left and right)** | Upper limbs | Left: thoracic duct. Right: right lymphatic duct |
| **Jugular (left and right)** | Head and neck | Left: thoracic duct. Right: right lymphatic duct |

### The two ducts

The right lymphatic duct is short and small. It drains the right upper limb and the right side of the head, neck, and thorax. The thoracic duct is the largest lymphatic vessel in the body. It drains everything else, which is about three quarters of the body. The thoracic duct begins in the abdomen as a sac called the cisterna chyli, which sits in front of the upper lumbar vertebrae. From

there it passes up through the diaphragm with the aorta and climbs through the thorax beside the vertebral column.

Both ducts end in the base of the neck. Each one empties into the junction of the subclavian vein and the internal jugular vein on its own side. The right lymphatic duct empties on the right side. The thoracic duct empties on the left side.

| Feature | Right lymphatic duct | Thoracic duct |
|---|---|---|
| **Size** | Short and small | Largest lymphatic vessel |
| **Drains** | Right upper limb and the right side of the head, neck, and thorax | Everything else, about three quarters of the body |
| **Trunks it receives** | Right jugular, right subclavian, right bronchomediastinal | Both lumbar, intestinal, left jugular, left subclavian, left bronchomediastinal |
| **Starts at** | The right side of the base of the neck | The cisterna chyli, a sac in the abdomen |
| **Empties into** | Junction of the right subclavian and right internal jugular veins | Junction of the left subclavian and left internal jugular veins |

- Memory tip: split the body into four quarters. The right upper quarter (right arm, right side of the head, neck, and chest) drains into the right lymphatic duct. The other three quarters drain into the thoracic duct. The right leg is thoracic duct territory, not right lymphatic duct territory.

### How lymph moves without a pump

The lymphatic system has no heart to push the lymph. Lymph is moved by forces from the structures around the vessels, and the valves keep it from sliding backward.

- Skeletal muscle pump: contracting muscles squeeze the vessels that run between them.
- Respiratory pump: pressure changes in the thorax with each breath pull lymph upward.
- Smooth muscle in the walls of the larger lymphatic vessels contracts in rhythm.
- Pulsing of nearby arteries presses on the vessels.
- Valves in the collecting vessels keep all of this movement going one way, toward the heart.

## Competency 106. Vessel and node structure

**You should be able to:** Describe the structure of lymphatic capillaries, valved collecting vessels, and a lymph node including capsule, cortex, medulla, hilum, and afferent and efferent vessels.

### Lymphatic capillaries

Lymphatic capillaries are the smallest lymphatic vessels. They lie in the tissues, woven between the blood capillaries. Each one is a thin tube that is closed at one end, so it is called blind-ended. Its wall is a single layer of endothelial cells.

The endothelial cells of a lymphatic capillary overlap at their edges. These loose, overlapping endothelial cells act as one-way flap valves. When fluid builds up in the tissue, the flaps push open and fluid flows in. When pressure inside the capillary rises, the flaps press shut, so the fluid cannot flow back out. Fine collagen strands called anchoring filaments tie the endothelial cells to the surrounding tissue. When the tissue swells, the filaments pull on the cells and hold the capillary open instead of letting it collapse.

Because the gaps between the cells are large, lymphatic capillaries are more permeable than blood capillaries. Proteins, cell debris, bacteria, and even cancer cells can get in. Lymphatic capillaries are found in nearly every tissue. They are absent from the central nervous system (the brain and spinal cord), bone and red bone marrow, and avascular tissues such as cartilage, the epidermis, and the cornea.

| Feature | Lymphatic capillary | Blood capillary |
|---|---|---|
| **Ends** | Blind (closed) at one end | Open at both ends, part of a loop |
| **Size** | Wider | Narrower |
| **Wall** | Overlapping endothelial cells act as one-way flap valves | Endothelial cells joined more tightly |
| **Held open by** | Anchoring filaments tied to the surrounding tissue | Pressure of the blood inside |
| **What gets in** | Fluid, proteins, cells, pathogens, and even cancer cells | Mostly small molecules |
| **Flow** | One way, away from the tissues | Through, from the artery side to the vein side |

- In the small intestine, each villus has a special lymphatic capillary in its center called a lacteal. Lacteals absorb dietary fats, which are packaged in particles too large to enter blood capillaries. The fat turns the lymph milky, and this lymph (chyle) drains to the cisterna chyli and up the thoracic duct.

### Collecting vessels and their valves

Lymphatic capillaries join to form lymphatic collecting vessels. These are built like small veins. Their walls have the same three layers as a vein, but the layers are thinner. Collecting vessels have more internal valves than veins do. These lymphatic valves are one-way flaps, so lymph can move forward but cannot flow backward. When a collecting vessel is full, the stretches between valves bulge, which gives the vessel a beaded look.

- Superficial collecting vessels run in the subcutaneous layer of the skin alongside the superficial veins. Deep collecting vessels run alongside the deep arteries.
- Along their length, the collecting vessels pass through lymph nodes.

### The lymph node

A lymph node is a small, bean-shaped organ that filters lymph. Hundreds of nodes lie along the collecting vessels. They are clustered in the neck (cervical nodes), the armpit (axillary nodes), and the groin (inguinal nodes), and there are many more deep in the thorax and abdomen.

The outside of a node is a capsule of dense connective tissue. Strands of connective tissue called trabeculae extend inward from the capsule and divide the node into compartments. Inside, a mesh of reticular fibers holds the lymphocytes and macrophages in place.

The node has two regions. The outer region is the cortex. The outer cortex holds rounded clusters of lymphocytes called lymphatic follicles, which are mostly B cells. Many follicles have a pale germinal center where B cells multiply. The deep cortex is mostly T cells. The inner region is the medulla. It contains medullary cords, strands of lymphocytes and plasma cells, separated by medullary sinuses, open channels lined with macrophages.

One side of the node is indented. This is the hilum. Blood vessels enter and leave the node at the hilum, and lymph leaves there too. Afferent lymphatic vessels carry lymph into the node. There are several of them, and they enter at many points on the rounded (convex) surface through the capsule. Efferent lymphatic vessels carry filtered lymph out of the node, and they leave only at the hilum. Because there are more afferent vessels than efferent vessels, lymph backs up and moves slowly through the node. This gives the macrophages and lymphocytes time to work.

| Structure | What it is |
|---|---|
| **Capsule** | Dense connective tissue covering; trabeculae extend inward from it |
| **Afferent lymphatic vessels** | Several vessels that carry lymph into the node through the capsule, on the convex surface |
| **Subcapsular sinus** | The space just under the capsule where lymph first enters |
| **Cortex** | Outer region; the outer cortex has lymphatic follicles of B cells with germinal centers, and the deep cortex is mostly T cells |
| **Medulla** | Inner region of medullary cords and medullary sinuses lined with macrophages |
| **Hilum** | The indented side of the node, where blood vessels enter and leave and efferent vessels leave |
| **Efferent lymphatic vessels** | Fewer vessels that carry filtered lymph out of the node at the hilum |

### The path of lymph through a node

1. Afferent lymphatic vessels: lymph enters through the capsule on the convex surface.
2. Subcapsular sinus: lymph spreads out in the space just under the capsule.
3. Cortical sinuses: lymph seeps through the cortex, past the follicles.
4. Medullary sinuses: macrophages lining these channels engulf bacteria, debris, and cancer cells.
5. Efferent lymphatic vessels: filtered lymph leaves at the hilum and moves on to the next node or a trunk.

### What keeps lymph moving one way at each step

| Structure | Wall or covering | What keeps lymph moving one way | Where lymph goes next |
|---|---|---|---|
| **Lymphatic capillary** | One layer of overlapping endothelial cells, held open by anchoring filaments | The overlapping cells act as flap valves | A collecting vessel |
| **Collecting vessel** | Built like a vein, with three thin layers | Many valves, more than veins have | A lymph node |
| **Lymph node** | A capsule of dense connective tissue | Lymph comes in only through the afferent vessels and leaves only through the efferent vessels | Out through the efferent vessels at the hilum, on toward a trunk |

- Memory tip: afferent vessels arrive, efferent vessels exit. The same two words mean "in" and "out" in other systems too, such as the arterioles of the kidney and the neurons of the nervous system.
- Afferent vessels enter all around the capsule, but efferent vessels leave only at the hilum. In a drawing, more should come in than go out.

## Competency 107. Primary and secondary organs

**You should be able to:** Sort red bone marrow, thymus, lymph nodes, spleen, tonsils, and MALT into primary and secondary lymphoid organs with location and role.

The main cells of the lymphatic system are lymphocytes, a type of white blood cell. The two kinds you need here are B cells and T cells. All lymphocytes are made in the red bone marrow. Before they can fight anything, they have to finish maturing. B cells

mature in the red bone marrow. T cell precursors leave the marrow and travel in the blood to the thymus, where they mature.

This gives a simple way to sort the lymphoid (lymphatic) organs. Primary lymphoid organs are where lymphocytes are made and mature. There are only two: the red bone marrow and the thymus. Secondary lymphoid organs are where mature lymphocytes gather and meet pathogens. These are the lymph nodes, the spleen, the tonsils, and MALT. Mature lymphocytes leave the marrow and thymus and travel out to the secondary organs. Each secondary organ sits where it can catch pathogens from one source: the lymph nodes check lymph, the spleen checks blood, and the tonsils and MALT guard the linings of open tracts.

| Organ | Group | Location | Role |
|---|---|---|---|
| **Red bone marrow** | Primary | Spongy bone of the axial skeleton (such as the sternum and vertebrae), the limb girdles (such as the hip bones), and the proximal ends of the femur and humerus | Where all lymphocytes form, and where B cells mature |
| **Thymus** | Primary | Mediastinum, above the heart and behind the sternum | Where T cells mature |
| **Lymph nodes** | Secondary | Along the collecting vessels, clustered in the cervical, axillary, and inguinal regions (neck, armpits, and groin) | Filter the lymph |
| **Spleen** | Secondary | Upper left abdomen, to the left of and behind the stomach, under the lower left ribs | Filters the blood |
| **Tonsils** | Secondary | A ring of lymphatic tissue in the walls of the pharynx and at the base of the tongue, around the entrance to the throat | Guard against inhaled and swallowed pathogens |
| **MALT** | Secondary | Mucous membranes of the digestive, respiratory, urinary, and reproductive tracts | Guards the mucosal surfaces |

- Memory tip: only two organs are primary, the red bone marrow and the thymus, because those are the only places lymphocytes are made and matured. B cells mature in the bone marrow and T cells mature in the thymus. Everything else on the list is secondary.
- The spleen is the largest lymphatic organ, but it is secondary. Size does not make an organ primary; making and maturing lymphocytes does.

### The thymus

The thymus sits in the mediastinum, above the heart and just behind the sternum. It is large in infants and children. After puberty it shrinks, and in older adults it is mostly fat. Its outer cortex is packed with immature T cells. Its inner medulla has fewer, more mature T cells and small round structures called thymic corpuscles. The thymus does not filter lymph. Unlike a lymph node, it has no afferent lymphatic vessels and no follicles.

### The spleen

The spleen is the largest lymphatic organ. It sits in the upper left abdomen against the diaphragm, to the left of and behind the stomach, and it is protected by the lower left ribs. The splenic artery and splenic vein pass through its hilum. The spleen filters the blood. It removes worn-out red blood cells and platelets and responds to pathogens carried in the blood. Its tissue comes in two kinds.

| Structure | What it is |
|---|---|
| **Spleen** | The largest lymphatic organ, in the upper left abdomen under the lower left ribs |
| **White pulp** | Small islands of lymphocytes clustered around central arteries; carries out the spleen's immune functions |
| **Red pulp** | Most of the organ; venous sinuses and splenic cords full of red blood cells and macrophages that filter the blood and remove worn-out erythrocytes |

### The tonsils and MALT

MALT stands for mucosa-associated lymphatic tissue. It is lymphatic tissue in the mucous membranes that line the open tracts of the body, where pathogens can get in. The largest and best organized pieces of MALT are the tonsils, the Peyer patches, and the appendix.

The tonsils form a ring of lymphatic tissue around the entrance to the pharynx, where they meet inhaled and swallowed material. Their surface has deep pits called crypts that trap bacteria and debris.

| Structure | Location | Notes |
|---|---|---|
| **Pharyngeal tonsil** | Posterior wall of the nasopharynx | Single; also called the adenoid |
| **Palatine tonsils** | Lateral walls of the oropharynx, between the palatoglossal and palatopharyngeal arches | Paired and the largest; the ones usually infected and removed |
| **Lingual tonsils** | Base (posterior third) of the tongue, facing the oropharynx | Guard swallowed material |
| **Tubal tonsils** | Around the openings of the auditory tubes in the nasopharynx | Paired |
| **Peyer patches** | Wall of the ileum, the last part of the small intestine | Clusters of lymphatic follicles that watch the intestinal contents |
| **Appendix** | Off the cecum of the large intestine | Wall packed with lymphatic follicles |

## Competency 108. Lymphatic disorders

**You should be able to:** Name edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis and identify the structure each one affects.

Lymphatic disorders fall into two groups. In the first group, drainage fails and fluid collects in the tissues, which shows up as swelling. In the second group, the lymphatic tissue itself changes. A node, the spleen, or the tonsils become enlarged, inflamed, or cancerous.

Edema is swelling caused by excess fluid in the interstitial space. It happens when fluid leaks out of the blood capillaries faster than the lymphatic vessels can carry it away. Lymphedema is a specific kind of edema. It is long-term swelling caused by blocked, damaged, or missing lymphatic vessels. It can follow surgery that removes lymph nodes, because the vessels that passed through those nodes no longer drain the region. The difference between the two is where the problem sits: edema is a problem in the tissue space, while lymphedema is a problem in the lymphatic vessels themselves.

Lymphadenopathy means enlarged lymph nodes. Nodes often swell when they are fighting an infection nearby, and they can also enlarge when cancer cells lodge in them. Lymphoma is a cancer of the lymphocytes in lymphatic tissue. It usually starts in lymph nodes and comes in two main groups, Hodgkin lymphoma and non-Hodgkin lymphoma. Lymphadenopathy is any enlarged node, for any reason; lymphoma is a cancer of the lymphocytes. Splenomegaly is an enlarged spleen. It can happen with an infection such as mononucleosis or with a blood disorder such as leukemia. Tonsillitis is inflammation of the tonsils, usually the palatine tonsils, and it is common in childhood.

| Disorder | Structure affected | What changes |
|---|---|---|
| **Edema** | The interstitial space | Excess interstitial fluid builds up and the tissue swells, because fluid leaks out faster than the lymphatics can drain it |
| **Lymphedema** | The lymphatic vessels | Blocked, damaged, or missing vessels cause long-term swelling of the region they drain, sometimes after lymph node removal |
| **Lymphadenopathy** | The lymph nodes | The nodes enlarge, often while fighting an infection, and sometimes because cancer cells have lodged in them |
| **Lymphoma** | Lymphatic tissue, usually the lymph nodes | Lymphocytes become cancerous and the tissue enlarges; Hodgkin or non-Hodgkin lymphoma |
| **Splenomegaly** | The spleen | The spleen enlarges, for example in mononucleosis or leukemia, and becomes easier to rupture |
| **Tonsillitis** | The tonsils, usually the palatine tonsils | The tonsils become infected, red, and swollen |

### What this means for patients

- Swollen glands are lymphadenopathy. Lymph nodes enlarge as they trap and fight pathogens, which is why a clinician feels for tender nodes in the neck, armpit, and groin during an exam.
- Because lymph passes through nodes in order, the lymphatic system also maps how some cancers spread. The first node a region drains into is the sentinel node, and checking it tells a surgeon whether a cancer has begun to travel. Removing many nodes can block drainage and leave lasting lymphedema in the limb.
- The spleen sits just under the lower left ribs. Its soft tissue tears easily, and it holds a large volume of blood, so a blow to that area can rupture the spleen and cause heavy internal bleeding. An enlarged spleen, as in mononucleosis, ruptures even more easily, which is why contact sports are dangerous during mononucleosis.

# The digestive system, part I: through the esophagus

The written notes for the first digestive lecture: the tube and its wall, the enteric plexuses, the peritoneum, the mouth, salivary glands, tongue, pharynx, esophagus, and the sphincters. Module 4 competencies 109 to 114, 120, and 121, with nine worked questions.

## Before the competencies: the two groups of organs and the six jobs

The digestive system is two groups of organs. The gastrointestinal (GI) tract, also called the alimentary canal, is one continuous tube from the mouth to the anus that runs through the thoracic and abdominal cavities: mouth, most of the pharynx, esophagus, stomach, small intestine, and large intestine. The accessory digestive organs sit beside the tube: the teeth and tongue, which touch food directly, and the salivary glands, liver, gallbladder, and pancreas, which make or store secretions and send them into the tube through ducts.

- In a living person the tract is about 5 to 7 m (16.5 to 23 feet) long, because its muscle is in a steady state of partial contraction (tonus). After death the muscle relaxes and the tube measures about 7 to 9 m (23 to 29.5 feet).

### The six basic processes of the GI tract

1. Ingestion: taking food and liquid into the mouth.
2. Secretion: water, acid, buffers, and enzymes released into the lumen.
3. Mixing and propulsion: the smooth muscle of the wall contracts and relaxes, mixing food with secretions and pushing it forward.
4. Digestion: breaking food into smaller molecules. Mechanical digestion is the cutting and grinding of the teeth and the churning of the stomach and small intestine. Chemical digestion is enzymes splitting carbohydrates, lipids, proteins, and nucleic acids by hydrolysis. Vitamins, ions, water, and cholesterol can be absorbed without chemical digestion.
5. Absorption: digested molecules cross the epithelium from the lumen into the blood or lymph.
6. Defecation: eliminating what cannot be used (feces: wastes, indigestible material, bacteria, and cells shed from the lining).

## Competency 109. Four-layer gut wall

**You should be able to:** Name the four layers of the alimentary canal wall from lumen outward and state the dominant tissue of each.

From the esophagus to the anal canal the wall of the alimentary canal has the same four layers. From the lumen outward: mucosa, submucosa, muscularis, serosa (or adventitia). Every region changes the details to suit its job, but the order never changes.

### The mucosa, the inner lining, has three parts

1. Epithelium, in direct contact with the food. Nonkeratinized stratified squamous epithelium in the mouth, pharynx, esophagus, and anal canal, where the job is protection from rubbing. Simple columnar epithelium in the stomach and intestines, where the job is secretion and absorption. Tight junctions seal the cells together so nothing leaks between them, and the cells turn over every 5 to 7 days. Exocrine cells secrete mucus and fluid into the lumen; enteroendocrine cells secrete hormones into the blood.
2. Lamina propria: areolar connective tissue carrying the blood and lymphatic vessels that pick up absorbed nutrients. It supports the epithelium and holds mucosa-associated lymphatic tissue (MALT), the lymphatic nodules that guard the tract. MALT is heaviest in the tonsils, the small intestine, the appendix, and the large intestine.
3. Muscularis mucosae: a thin layer of smooth muscle. In the stomach and small intestine it throws the lining into small folds, which adds surface area and keeps every cell exposed to the contents.

### The other three layers

| Layer | Main tissue | What it holds |
|---|---|---|
| **Submucosa** | Areolar connective tissue (OpenStax calls it dense); binds the mucosa to the muscularis | Larger blood and lymphatic vessels that receive absorbed food molecules; the submucosal plexus; glands in some regions |
| **Muscularis** | Smooth muscle in most of the tube, in two sheets: an inner circular layer and an outer longitudinal layer, with the myenteric plexus between them | Skeletal muscle instead in the mouth, pharynx, and the upper and middle esophagus (voluntary swallowing) and in the external anal sphincter (voluntary defecation) |
| **Serosa** | A serous membrane: simple squamous epithelium (mesothelium) over areolar connective tissue. It is the visceral peritoneum and suspends the tract in the abdominal cavity | Where the tube is not in the peritoneal cavity, an adventitia of areolar connective tissue replaces it: most of the esophagus, and the back of retroperitoneal organs |

- Read the wall from the inside out: mucosa, submucosa, muscularis, serosa.
- The muscularis mucosae is a thin layer inside the mucosa. It is not the muscularis. Keep them in separate rows of your table.
- Two places break the muscle pattern. The stomach has a third, oblique layer (competency 115). The large intestine gathers its outer longitudinal layer into three bands, the teniae coli (competency 117).

## Competency 110. Enteric nerve plexuses

**You should be able to:** Locate the submucosal and myenteric plexuses in the gut wall and match each to the wall layer it sits within.

The tract has its own nervous system built into its wall, the enteric nervous system, sometimes called the brain of the gut. About 100 million neurons run from the esophagus to the anus, arranged in two plexuses. Each sits in a different layer, and the layer tells you the job.

| Plexus | Wall layer | Motor neurons supply | Controls |
|---|---|---|---|
| **Myenteric plexus (Auerbach)** | Muscularis, between the inner circular and outer longitudinal layers | The two smooth muscle layers | Motility: how often and how hard the muscularis contracts |
| **Submucosal plexus (Meissner)** | Submucosa, just deep to the mucosa | The secretory cells of the mucosa and the vessels to it | Secretion by the GI organs, and blood flow to the lining |

- Each plexus holds motor neurons, interneurons, and sensory neurons. Interneurons connect the two plexuses. Sensory neurons reach the mucosa, where chemoreceptors respond to chemicals in the food and mechanoreceptors respond to stretch when the wall fills.
- The name tells you the layer. Sub-mucosal is in the submucosa. Myenteric comes from myo, muscle, so it sits in the muscularis.
- Secretion goes with the submucosal plexus; movement goes with the myenteric plexus.

### The autonomic nervous system adjusts the enteric system

- Parasympathetic fibers reach the tract through the vagus nerve (cranial nerve X) for everything down to about the first half of the large intestine, and through pelvic splanchnic nerves from the sacral cord for the last half. Preganglionic fibers synapse on neurons in the myenteric and submucosal plexuses; some reach the smooth muscle and glands directly. Parasympathetic activity increases secretion and motility.
- Sympathetic fibers come from the thoracic and upper lumbar cord. Postganglionic fibers synapse on plexus neurons and inhibit them. Sympathetic activity decreases secretion and motility, which is why anger, fear, and anxiety slow digestion.
- GI reflexes: receptors in the wall sense stretch or chemicals, signal neurons in the enteric system, the CNS, or the ANS, and those neurons turn GI glands and smooth muscle up or down.

## Competency 111. Peritoneum and folds

**You should be able to:** Identify the parietal and visceral peritoneum, the peritoneal cavity, and the five folds greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon and state what each fold connects.

The peritoneum is the largest serous membrane in the body: a layer of simple squamous epithelium (mesothelium) on areolar connective tissue. Like the pleura, it has two layers with a film of serous fluid between them so the organs slide without friction.

**Parietal peritoneum**
Lines the walls of the abdominopelvic cavity.

**Visceral peritoneum**
Covers the organs. On a canal organ it is the serosa.

**Peritoneal cavity**
The thin space between the two layers, holding a little lubricating peritoneal fluid.

**Ascites**
In liver disease, pancreatic disease, and some cancers, the peritoneal cavity fills with liters of fluid that often has to be drained.

Unlike the pleura, the peritoneum has large folds that weave between the organs. A fold is a double layer of peritoneum. The folds hold organs to each other and to the walls, store fat between their layers, and carry the blood vessels, lymphatic vessels, and nerves to the organs. Five are named in this course. For each one, learn its two ends.

| Fold | What it connects | Other features |
|---|---|---|
| **Greater omentum** | Hangs from the greater curvature of the stomach and the first part of the duodenum, drapes down in front of the small intestine, then folds back up and attaches to the transverse colon | The largest fold. Folds on itself into four layers. A fatty apron that grows with weight gain and holds many lymph nodes that fight infections of the tract |
| **Falciform ligament** | Attaches the liver to the anterior abdominal wall and the diaphragm | The liver is the only organ attached to the anterior wall. Its free lower edge holds the ligamentum teres |
| **Lesser omentum** | From the lesser curvature of the stomach and the first part of the duodenum up to the liver | Its free edge carries the hepatic portal vein, the hepatic artery, and the common bile duct, with some lymph nodes |
| **Mesentery** | A fan from the posterior abdominal wall to the jejunum and ileum | Double-layered, full of fat, with the vessels, lymphatics, and lymph nodes of the small intestine between its layers |
| **Mesocolon** | Two separate folds from the posterior wall: the transverse mesocolon to the transverse colon and the sigmoid mesocolon to the sigmoid colon | Carries the vessels to those parts of the colon. With the mesentery, it holds the intestines loosely so they can move as they mix and push food |

- Greater goes with the greater curvature and hangs low. Lesser goes with the lesser curvature and runs up to the liver. Meso means middle: the mesentery and mesocolon hold the intestines out from the back wall like a hammock.
- The falciform ligament reaches the front wall; the mesentery and mesocolon reach the back wall.
- On the cadaver, the greater omentum is the first thing you see when the abdomen is opened. Lift it toward the head to find the transverse colon on its back surface and the small intestine below.

### What this means for patients

- Peritonitis is acute inflammation of the peritoneum. Microbes get in through a surgical or accidental wound, a ruptured appendix, or a perforated intestine, and the infection can be life-threatening.
- Inflamed peritoneal surfaces rubbing together is also peritonitis, painful but not usually dangerous. People on peritoneal dialysis are at risk.

## Competency 112. Retroperitoneal organs

**You should be able to:** Distinguish retroperitoneal organs from intraperitoneal ones and name the kidneys, pancreas, duodenum, and parts of the colon as retroperitoneal.

**Intraperitoneal organ**
Wrapped almost completely in visceral peritoneum and hanging into the cavity on a fold, so it can move: the stomach, jejunum, ileum, transverse colon, sigmoid colon, liver, and spleen.

**Retroperitoneal organ**
Lies on the posterior abdominal wall in the retroperitoneum, the region behind the parietal peritoneum, with peritoneum on its front surface only. It is not in the peritoneal cavity, has no mesentery, and is held firmly in place. Its back surface is covered by an adventitia.

| Organ or part | Position | How you can tell |
|---|---|---|
| **Stomach** | Intraperitoneal | Hangs between the lesser and greater omenta |
| **Duodenum** | Retroperitoneal (except its first part) | Fixed to the back wall in a C around the head of the pancreas |
| **Jejunum and ileum** | Intraperitoneal | Hang on the mesentery |
| **Ascending colon** | Retroperitoneal | Fixed to the right posterior wall |
| **Transverse colon** | Intraperitoneal | Hangs on the transverse mesocolon |
| **Descending colon** | Retroperitoneal | Fixed to the left posterior wall |
| **Sigmoid colon** | Intraperitoneal | Hangs on the sigmoid mesocolon |
| **Pancreas** | Retroperitoneal (except the tip of the tail) | Lies across the back wall behind the stomach |
| **Kidneys** | Retroperitoneal | Against the back wall on either side of the spine |

- The colon alternates: ascending fixed, transverse hanging, descending fixed, sigmoid hanging. The two parts with a mesocolon are the two intraperitoneal parts.
- SADPUCKER for the long list: Suprarenal (adrenal) glands, Aorta and inferior vena cava, Duodenum (except the first part), Pancreas (except the tail), Ureters, Colon (ascending and descending), Kidneys, Esophagus (the abdominal part), Rectum.

## Competency 113. Mouth, pharynx, esophagus

**You should be able to:** Identify the oral cavity boundaries, hard and soft palate, uvula, fauces, tonsils, the three pharyngeal regions, and the esophagus with its hiatus.

### The mouth (oral cavity, buccal cavity)

| Boundary | What forms it |
|---|---|
| **Anterior** | The lips (labia): fleshy folds of skin outside and nonkeratinized stratified squamous mucosa inside, with the orbicularis oris between. The labial frenulum, a midline fold, ties each lip to its gum |
| **Lateral** | The cheeks: skin outside, mucosa inside, the buccinator muscle between. Lips and cheeks keep food between the teeth and help with speech |
| **Superior (roof)** | The hard palate in front and the soft palate behind |
| **Inferior (floor)** | The tongue and the muscles under it (mylohyoid) |
| **Posterior** | The fauces, the opening into the oropharynx |

**Oral vestibule**
The space between the lips and cheeks on the outside and the gums and teeth on the inside.

**Oral cavity proper**
The space from the gums and teeth back to the fauces.

**Hard palate**
The bony front of the roof, formed by the palatine processes of the maxillae and the horizontal plates of the palatine bones under mucosa. It walls the mouth off from the nasal cavity, so you can chew and breathe at the same time, and gives the tongue a firm surface to press food against.

**Soft palate**
The muscular, arch-shaped back of the roof, between the oropharynx and nasopharynx. Its back border opens into the oropharynx through the fauces.

**Uvula**
The finger-shaped muscular tag hanging from the back edge of the soft palate. During a swallow the soft palate and uvula lift and seal the nasopharynx, so food and liquid do not enter the nose.

**Palatoglossal arch**
The front arch, running from the soft palate down to the base of the tongue.

**Palatopharyngeal arch**
The back arch, running from the soft palate to the side of the pharynx.

**Palatine tonsils**
Between the two arches on each side. These are the tonsils you see in an open mouth.

**Lingual tonsil**
On the back (root) of the tongue.

### The pharynx

A funnel-shaped tube of skeletal muscle lined with mucous membrane, from the choanae to the esophagus behind and the larynx in front. Three regions: the nasopharynx (air only), the oropharynx, and the laryngopharynx (air and food). The pharyngeal tonsil sits in the nasopharynx. A swallowed bolus goes mouth, oropharynx, laryngopharynx, esophagus, stomach, pushed by the pharyngeal constrictors. Breathing stops for the moment the bolus passes.

### The esophagus

- A collapsible muscular tube about 25 cm (10 inches) long. It starts at the lower end of the laryngopharynx at the level of the cricoid cartilage, runs through the mediastinum behind the trachea and in front of the vertebral column, pierces the diaphragm, and joins the stomach at the cardia.
- The esophageal hiatus is the opening in the diaphragm it passes through, at about T10.
- It secretes mucus and carries food to the stomach. No digestion and no absorption happen here.

| Layer | In the esophagus |
|---|---|
| **Mucosa** | Nonkeratinized stratified squamous epithelium (wear and tear), lamina propria, muscularis mucosae. Mucous glands near the stomach |
| **Submucosa** | Areolar connective tissue, vessels, mucous glands |
| **Muscularis** | Upper third skeletal muscle, middle third mixed, lower third smooth muscle. Two sphincters: the upper esophageal sphincter (skeletal) and the lower esophageal (cardiac) sphincter (smooth), near the heart |
| **Adventitia** | Areolar connective tissue with no mesothelium, blending into the mediastinum to anchor the esophagus. Only the short abdominal segment has a serosa |

### What this means for patients

- Hiatal hernia: part of the stomach pushes up through the esophageal hiatus into the thorax. It weakens the support the diaphragm gives the lower esophageal sphincter and often comes with reflux (competency 119).

## Competency 114. Major canal sphincters

**You should be able to:** Locate the upper and lower esophageal, pyloric, ileocecal, and internal and external anal sphincters and state which regions each one separates.

A sphincter is a ring of muscle around the tube that can squeeze it shut. Sphincters sit at borders between regions. They control when contents move on and most keep contents from flowing backward. Learn each one by the two regions on either side of it.

| Sphincter | Where | Separates | Muscle |
|---|---|---|---|
| **Upper esophageal sphincter** | Top of the esophagus, at the cricoid cartilage | Laryngopharynx from esophagus | Skeletal; stays closed between swallows so air does not enter the esophagus |
| **Lower esophageal sphincter (also called the cardiac sphincter or gastroesophageal sphincter)** | Bottom of the esophagus, just under the diaphragm | Esophagus from stomach (cardia) | Smooth; the diaphragm around the hiatus helps hold it shut, keeping acid out of the esophagus |
| **Pyloric sphincter** | Around the pylorus at the stomach exit | Stomach from duodenum | Smooth, a thickening of the circular layer; meters small amounts of chyme into the duodenum |
| **Ileocecal sphincter** | End of the ileum where it opens into the cecum | Small intestine (ileum) from large intestine (cecum) | Smooth; with folds of mucosa it forms the ileocecal valve, which keeps colon contents out of the ileum |
| **Internal anal sphincter** | Around the anal canal, inside the external sphincter | Anal canal from the outside | Smooth, involuntary |
| **External anal sphincter** | Around the anal canal, outside the internal sphincter | Anal canal from the outside | Skeletal, voluntary |

- Only the two ends of the canal have skeletal sphincters: the upper esophageal at the top and the external anal at the bottom. Everything in between is smooth muscle.
- The sphincter of Oddi (hepatopancreatic sphincter) is on your lab list with the small intestine, but it is not a sphincter of the canal. It controls bile and pancreatic juice entering the duodenum at the major duodenal papilla (competencies 123 and 124).

## Competency 121. Salivary glands

**You should be able to:** Locate the parotid, submandibular, and sublingual glands and state where the duct of each opens.

Saliva keeps the mouth and pharynx moist, cleans the mouth and teeth, and during a meal lubricates and dissolves food and starts the chemical breakdown of starch. Small glands in the lining (labial, buccal, palatal, and lingual glands) keep the mouth wet all the time. Most saliva comes from three pairs of major glands outside the lining, each sending its saliva in through a duct.

| Gland | Location | Where the duct opens | Saliva |
|---|---|---|---|
| **Parotid gland** | In front of and below the ear, between the skin and the masseter muscle; the largest | The parotid duct crosses the masseter, pierces the buccinator, and opens into the vestibule opposite the upper second molar | Watery (serous), with salivary amylase |
| **Submandibular gland** | Floor of the mouth, along the inner surface of the mandible | The submandibular duct runs forward under the mucosa of the floor and opens just lateral to the lingual frenulum | Mostly serous with some mucus, a little thicker |
| **Sublingual gland** | Under the tongue in the floor of the mouth, above the submandibular gland; the smallest | Many small sublingual ducts opening along the floor of the mouth | Mostly mucus, thick, very little amylase |

- Each name tells you the spot. Parotid, beside the ear. Submandibular, below the mandible. Sublingual, below the tongue.
- Only the parotid duct opens in the cheek. The other two open under the tongue.
- Saliva is about 99.5 percent water. The rest is ions (sodium, potassium, chloride, bicarbonate, phosphate), dissolved gases, and organic substances: urea, uric acid, mucus, IgA, lysozyme, and salivary amylase, which splits starch into maltose, maltotriose, and alpha-dextrins.

### What this means for patients

- Mumps is a paramyxovirus infection that targets the parotid glands. The glands swell on one or both sides in front of the ramus of the mandible, with fever, malaise, and throat pain on swallowing or with sour foods. In about 30 percent of males past puberty the testes become inflamed; sterility is rare unless both sides are involved. A vaccine prevents it.

## Competency 120. Teeth and tongue

**You should be able to:** Identify the teeth and tongue with the lingual frenulum, papillae, intrinsic and extrinsic muscles, and lingual glands.

### The tongue

The tongue is skeletal muscle covered with mucous membrane. With its muscles it forms the floor of the oral cavity. A median septum divides it into right and left halves along its whole length, and it attaches below to the hyoid bone, the styloid process of the temporal bone, and the mandible.

| Muscle group | Origin and insertion | Action |
|---|---|---|
| **Extrinsic (genioglossus, hyoglossus, styloglossus, palatoglossus)** | Start on bone or the soft palate outside the tongue, end in the connective tissue of the tongue | Move the whole tongue: side to side, in and out, up and down, to position food for chewing, shape the bolus, and push it back to swallow. They also hold the tongue in place |
| **Intrinsic (superior and inferior longitudinal, transverse, vertical)** | Start and end inside the tongue | Change the shape and size of the tongue for speech and swallowing: curl, flatten, narrow, shorten |

- The hypoglossal nerve (cranial nerve XII) supplies all the tongue muscles except palatoglossus, which the vagus supplies.
- Extrinsic starts outside the tongue and moves it. Intrinsic sits within it and changes its shape.

**Lingual frenulum**
The midline fold of mucosa under the tongue that ties it to the floor of the mouth and limits how far back it can move. A short or rigid frenulum is ankyloglossia (tongue-tie), which impairs speech and can be corrected surgically.

**Lingual papillae**
Projections of the lamina propria covered with stratified squamous epithelium on the top and sides of the tongue. Some carry taste buds; all give the tongue grip on food.

**Lingual glands**
Small glands in the lamina propria of the tongue. They release mucus and a watery fluid with lingual lipase, which starts the breakdown of dietary triglycerides into fatty acids and diglycerides.

| Papillae | Where and what they look like | Taste buds |
|---|---|---|
| **Filiform** | Thin, pointed, the most numerous, over most of the top surface | No. They give the tongue friction |
| **Fungiform** | Mushroom-shaped, scattered among the filiform, most on the tip and sides | Yes |
| **Vallate (circumvallate)** | Large, round, about 8 to 12 in a V-shaped row near the back, each in a trench | Yes |
| **Foliate** | Leaf-like folds on the sides toward the back | Yes |

### The teeth

- Each tooth sits in a socket (an alveolus) in the maxilla or mandible, held by a periodontal ligament. The joint is a gomphosis, a fibrous joint.
- Children have 20 deciduous teeth. Adults have 32 permanent teeth: 8 incisors (cut), 4 canines (tear), 8 premolars (crush), and 12 molars (grind), including the four third molars (wisdom teeth).

**Crown, neck, root**
The crown is above the gum (gingiva), the neck is at the gum line, and the root sits in the socket.

**Enamel**
Covers the crown. The hardest substance in the body.

**Dentin**
Forms most of the tooth, under the enamel.

**Pulp cavity and root canal**
The center of the tooth, holding the pulp (vessels, nerves, connective tissue). It continues down each root as the root canal.

**Cementum**
Covers the root and anchors it to the periodontal ligament.

# The digestive system, part II: stomach, pancreas, liver, and gallbladder

The written notes for the second digestive lecture: the stomach, the pancreas, the liver and its lobule, the gallbladder and the path of bile, and the accessory organ disorders. Module 4 competencies 115 and 122 to 125, with nine worked questions.

## Competency 115. Stomach regions and features

**You should be able to:** Identify the cardia, fundus, body, pyloric part, curvatures, rugae, and the three-layered muscularis of the stomach.

The stomach is a J-shaped widening of the canal just under the diaphragm, connecting the esophagus above to the duodenum below. It is a mixing chamber and a holding tank: it stores the meal, churns it with gastric juice until the semisolid bolus is a liquid called chyme, begins the digestion of starch, triglycerides, and proteins, absorbs a few substances, and releases chyme into the duodenum a little at a time. Empty, it is about the size of a large sausage. Full, it stretches to hold 2 to 3 liters.

### The four regions

| Region | Where it is |
|---|---|
| **Cardia (cardiac region)** | The small region around the opening from the esophagus |
| **Fundus** | The rounded dome that bulges above and to the left of the cardia, up under the diaphragm |
| **Body** | The large central part below the fundus |
| **Pyloric part (pyloric region)** | The funnel that connects the body to the duodenum, in three parts in order: the pyloric antrum (wide, joins the body), the pyloric canal (narrow), and the pylorus (the outlet, surrounded by the pyloric sphincter) |

**Lesser curvature**
The short, concave border on the right (medial) side, facing the liver. The lesser omentum attaches here and runs up to the liver.

**Greater curvature**
The long, convex border on the left (lateral) side. The greater omentum hangs from here and drapes over the intestines.

**Rugae**
Large folds of mucosa and submucosa on the inside of an empty stomach. They flatten as the stomach fills, which is how it stretches.

**Pyloric sphincter**
The ring of smooth muscle around the pylorus that meters chyme into the duodenum. A thickening of the circular layer.

- Draw the fundus up beside the esophagus, not at the bottom of the J. Food collects there, but it does not have to pass through it to move on.
- Our notes call the curvatures medial and lateral (OpenStax); your slides call them superior and inferior. Both describe the same two borders.

### The wall of the stomach

| Layer | In the stomach |
|---|---|
| **Mucosa** | Simple columnar epithelium of surface mucous cells. The surface is dotted with gastric pits, narrow channels that gastric glands open into. Lamina propria of areolar connective tissue; muscularis mucosae of smooth muscle |
| **Submucosa** | Areolar connective tissue |
| **Muscularis** | Three layers of smooth muscle instead of two: inner oblique (the extra layer, mostly in the body), middle circular (thickens into the pyloric sphincter), outer longitudinal. The oblique fibers let the stomach twist and squeeze as well as push |
| **Serosa** | Visceral peritoneum. At the lesser curvature it continues up to the liver as the lesser omentum; at the greater curvature it continues down as the greater omentum |

### The cells of the gastric glands

| Cell | Secretes |
|---|---|
| **Mucous neck cells** | Mucus |
| **Chief cells** | Pepsinogen (the inactive form of the protein-digesting enzyme pepsin) and gastric lipase |
| **Parietal cells** | Hydrochloric acid, and intrinsic factor, which is needed to absorb vitamin B12 |
| **G cells (enteroendocrine), mostly in the pyloric antrum** | Gastrin, a hormone that stimulates gastric activity |

Together the exocrine secretions are gastric juice.

**What this means for patients**

- Pylorospasm: in some infants the pyloric sphincter fails to relax, food cannot leave the stomach, the stomach overfills, and the baby vomits to relieve the pressure. Treated with drugs that relax the sphincter.
- Pyloric stenosis: a narrowing of the pyloric sphincter in infants. The hallmark is projectile vomiting. It needs surgical correction.
- Vomiting (emesis) is the forceful expulsion of the contents of the stomach, and sometimes the duodenum, through the mouth. Irritation or stretch of the stomach, unpleasant sights, general anesthesia, dizziness, and drugs such as morphine and digitalis all signal the vomiting center in the medulla (the area postrema). The diaphragm and abdominal muscles squeeze the stomach and the contents come up through relaxed esophageal sphincters. Prolonged vomiting in infants and the elderly is dangerous: loss of acidic gastric juice shifts blood pH toward alkalosis, the person dehydrates, and acid damages the esophagus and teeth.

## Competency 124. Pancreas structure

**You should be able to:** Identify the head, body, and tail of the pancreas with its main and accessory ducts, major duodenal papilla, sphincter of Oddi, acini, and islets.

Most chemical digestion happens in the small intestine, and the pancreas, liver, and gallbladder supply the tools. The pancreas is a long, soft gland lying across the posterior abdominal wall behind the greater curvature of the stomach. It is retroperitoneal. It is an exocrine gland (pancreatic juice down ducts into the duodenum) and an endocrine gland (hormones into the blood).

| Structure | What it is |
|---|---|
| **Head** | The wide end, in the C-shaped curve of the duodenum, right of the midline |
| **Body** | The middle part, crossing the midline behind the stomach, in front of the aorta and the vertebral column |
| **Tail** | The narrow left end, near the spleen |
| **Main pancreatic duct (duct of Wirsung)** | The larger duct. Runs the whole length of the gland from tail to head, collecting pancreatic juice. In the wall of the duodenum it joins the common bile duct at the hepatopancreatic ampulla (ampulla of Vater) |
| **Accessory pancreatic duct (duct of Santorini)** | The smaller duct. Drains part of the head and opens on its own into the duodenum at the minor duodenal papilla, a little above the major papilla |
| **Hepatopancreatic ampulla** | The short widened chamber where the common bile duct and main pancreatic duct meet |
| **Major duodenal papilla** | The bump in the duodenal lining where the ampulla opens, in the descending part of the duodenum below the pyloric sphincter |
| **Sphincter of Oddi (hepatopancreatic sphincter)** | The ring of smooth muscle around the ampulla that controls bile and pancreatic juice flowing into the duodenum |
| **Acini** | Small clusters of gland cells around tiny ducts: the exocrine part, about 99 percent of the gland. They make pancreatic juice, 1200 to 1500 mL a day of clear fluid: water, salts, sodium bicarbonate, and digestive enzymes |
| **Pancreatic islets (islets of Langerhans)** | Pale round patches scattered among the acini: the endocrine part, about 1 percent. They release glucagon, insulin, somatostatin, and pancreatic polypeptide into the blood |

- Match the size words. The main duct opens at the major papilla. The accessory duct opens at the minor papilla.
- Only the main duct joins the common bile duct. The accessory duct empties on its own.

## Competency 122. Liver gross and lobule anatomy

**You should be able to:** Identify the four liver lobes, falciform ligament, and ligamentum teres, and name the hepatocytes, sinusoids, canaliculi, and portal triad of the lobule.

The liver is the largest gland in the body. It sits under the diaphragm, filling most of the right hypochondriac region and part of the epigastric region, protected by the lower ribs. Visceral peritoneum covers most of it, and under that a capsule of dense irregular connective tissue covers all of it.

**Lobes and ligaments**

| Structure | Where and what |
|---|---|
| **Right lobe** | The largest lobe, right of the falciform ligament |
| **Left lobe** | Left of the falciform ligament, reaching over the stomach |
| **Quadrate lobe** | A small square lobe on the inferior surface, between the gallbladder and the ligamentum teres |
| **Caudate lobe** | A small lobe on the posterior surface beside the inferior vena cava |
| **Porta hepatis** | The doorway between the quadrate and caudate lobes. The hepatic portal vein and hepatic artery proper enter here; the right and left hepatic ducts leave here |
| **Falciform ligament** | The sickle-shaped fold of peritoneum that suspends the liver from the diaphragm and the anterior abdominal wall and marks the line between the right and left lobes on the front |
| **Ligamentum teres (round ligament)** | The fibrous cord in the free edge of the falciform ligament, running from the umbilicus to the liver. It is what remains of the fetal umbilical vein |
| **Coronary ligaments** | Narrow extensions of parietal peritoneum that hang the liver from the diaphragm |
| **Lesser omentum** | The fold from the lesser curvature of the stomach and the start of the duodenum to the visceral surface of the liver. Its free edge carries the portal vein, hepatic artery proper, and common bile duct |

- From the front you see only two lobes. The quadrate and caudate lobes are underneath and behind.
- The ligamentum teres comes from the umbilical vein, which carried oxygen-rich blood from the placenta. Vein, not artery.

### The liver lobule

Liver tissue is thousands of small six-sided units. Hepatocytes are the working cells: specialized epithelial cells with 5 to 12 sides making up about 80 percent of the liver's volume. They are arranged in plates one cell thick, the hepatic laminae, that radiate out from a central vein like spokes. Between the plates run the sinusoids.

| Structure | Where it is | What it does |
|---|---|---|
| **Central vein** | The center of the lobule | Collects blood from the sinusoids |
| **Hepatocytes** | In plates (hepatic laminae) radiating from the central vein | The working cells: they make bile and do the liver's metabolic work |
| **Hepatic sinusoids** | Between the plates | Wide, leaky, endothelium-lined capillaries carrying mixed blood from the edge of the lobule toward the central vein. Stellate macrophages (Kupffer cells) line them and destroy worn-out blood cells, bacteria, and foreign matter arriving from the gut |
| **Bile canaliculi** | Grooves between neighboring hepatocytes inside each plate | Collect the bile the hepatocytes make and carry it outward to the edge of the lobule |
| **Portal triad** | At each corner of the lobule | A branch of the hepatic portal vein, a branch of the hepatic artery, and a bile duct. The two vessels bring blood in; the duct carries bile out |

- Blood flows in toward the central vein. Bile flows out toward the portal triad. Opposite directions.
- Bile is a yellow-brown, sometimes olive-green liquid. It is both an excretory product and a digestive secretion.

### Blood through the liver, in order

1. The hepatic artery proper (oxygen-rich blood from the aorta) and the hepatic portal vein (oxygen-poor blood carrying newly absorbed nutrients, drugs, microbes, and toxins from the gut and spleen) enter at the porta hepatis.
2. Branches of both run to the portal triads at the corners of every lobule.
3. The two streams mix in the sinusoids and flow past the hepatocytes toward the center.
4. The central vein collects the blood.
5. Central veins join into hepatic veins, which leave the back of the liver.
6. The hepatic veins empty into the inferior vena cava just under the diaphragm, and the blood returns to the right atrium.

Because so much blood from the gut passes through the liver, it is a very common site for cancers of the digestive tract to spread.

### What the liver does

- Makes bile and the bile salts in it, which emulsify fats in the small intestine so they can be absorbed.
- Carbohydrate metabolism: keeps blood glucose steady and stores extra glucose as glycogen.
- Lipid metabolism and protein metabolism, including removing the amino group from amino acids (deamination) and sending the ammonia out as urea in the urine.
- Processes drugs and hormones, including alcohol and antibiotics.
- Excretes bilirubin, the pigment left over from breaking down old red blood cells.
- Stores glycogen and vitamins A, B12, D, E, and K.
- Phagocytosis by the Kupffer cells, and the final activation of vitamin D.

## Competency 123. Gallbladder and bile pathway

**You should be able to:** Identify the gallbladder regions and trace bile from the canaliculi through the hepatic, cystic, and common bile ducts to the hepatopancreatic ampulla and duodenum.

The gallbladder is a pear-shaped muscular sac about 8 to 10 cm long, sitting in a shallow hollow on the visceral surface of the right lobe of the liver and hanging a little past its lower edge. It stores bile between meals and concentrates it by removing water. When fatty food reaches the duodenum, the smooth muscle in its wall contracts and squeezes bile out.

| Structure | What it is |
|---|---|
| **Fundus** | The wide, rounded end that sticks out past the lower border of the liver |
| **Body** | The main middle part |
| **Neck** | The narrow end that points toward the porta hepatis and continues as the cystic duct |
| **Right and left hepatic ducts** | Drain bile from the two sides of the liver and leave at the porta hepatis |
| **Common hepatic duct** | Formed where the right and left hepatic ducts join |
| **Cystic duct** | Runs from the neck of the gallbladder and joins the common hepatic duct. Carries bile in both directions |
| **Common bile duct** | Formed where the common hepatic duct and the cystic duct join. Runs down in the edge of the lesser omentum, behind the first part of the duodenum, and through the head of the pancreas |
| **Hepatopancreatic ampulla, sphincter of Oddi, major duodenal papilla** | Where the common bile duct meets the main pancreatic duct and empties into the descending duodenum (competency 124) |

### A drop of bile, from the liver to the duodenum

1. Hepatocytes make the bile.
2. Bile canaliculi between the hepatocytes collect it and carry it to the edge of the lobule.
3. Bile ductules and the bile ducts of the portal triads carry it out of the lobules and join into larger ducts.
4. Right and left hepatic ducts leave the liver at the porta hepatis.
5. They join to form the common hepatic duct.
6. Side trip: when the sphincter of Oddi is closed, bile backs up and turns up the cystic duct into the gallbladder for storage. When the gallbladder contracts, bile comes back down the same cystic duct.
7. Common hepatic duct plus cystic duct form the common bile duct.
8. At the hepatopancreatic ampulla the common bile duct joins the main pancreatic duct.
9. The sphincter of Oddi opens and bile enters the duodenum through the major duodenal papilla, where it mixes with food.

- The duct names tell you where each duct comes from. Hepatic means liver. Cystic means bladder, this bladder. The common bile duct is where both come together.
- The cystic duct joins the common hepatic duct. It does not run straight to the duodenum. The gallbladder is a side trip, not a step every drop takes.

## Competency 125. Clinical anatomy of accessory disorders

**You should be able to:** Match gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps to the accessory organ each one affects and explain how a stone at the ampulla can block bile and pancreatic flow at once.

| Disorder | Organ affected | What changes |
|---|---|---|
| **Gallstones** | Gallbladder and bile ducts | Hard deposits form from bile in the gallbladder. A stone can move into the cystic duct or the common bile duct and block it |
| **Cholecystitis** | Gallbladder | The wall is inflamed and swollen, most often because a stone blocks the cystic duct and traps bile inside |
| **Hepatitis** | Liver | The liver is inflamed and hepatocytes are damaged, most often by a virus, also by alcohol or some drugs |
| **Cirrhosis** | Liver | After long-term damage, hepatocytes are replaced by bands of scar tissue and lumps of regrowing cells. The lobules lose their pattern and blood has trouble getting through |
| **Jaundice** | Liver or bile ducts, seen in the skin and eyes | Bilirubin builds up in the blood and stains the skin and the whites of the eyes yellow, because the liver cannot clear it into bile, a duct is blocked, or red cells break down too fast |
| **Pancreatitis** | Pancreas | The pancreas is inflamed and swollen. Digestive enzymes become active inside the gland and digest its own tissue. Gallstones and alcohol are common causes |
| **Mumps** | Parotid glands | A viral infection swells the parotids, so the face puffs out in front of and below the ears |

**Shared ducts, shared problems**

- The common bile duct and the main pancreatic duct meet at the hepatopancreatic ampulla and share one exit into the duodenum. A gallstone that travels down the common bile duct and lodges in the ampulla plugs that shared exit.
- Bile backs up into the liver and the blood: jaundice.
- Pancreatic juice backs up into the pancreas, where its enzymes damage the gland: pancreatitis.
- One stone, two organs. That is one way gallstones lead to pancreatitis.
- Mumps swells the parotid gland, not the pancreas. Cholecystitis is the gallbladder; hepatitis and cirrhosis are the liver.

# The digestive system, part III: small intestine, large intestine, and rectum

The written notes for the third digestive lecture: the small intestine and its three levels of folding, the large intestine, the rectum and anal canal, the whole path of food, and the canal disorders. Module 4 competencies 116 to 119, with nine worked questions.

## Competency 116. Small intestine and surface area

**You should be able to:** Identify the duodenum, jejunum, and ileum and name the circular folds, villi, lacteal, and microvilli as features that expand absorptive surface area.

The small intestine runs from the pyloric sphincter to the ileocecal valve. It is the longest part of the canal, about 3 m (10 feet) in a living person, and fills most of the middle and lower abdomen. Most chemical digestion and nearly all absorption of nutrients happen here.

| Region | Length | Location and features |
|---|---|---|
| **Duodenum** | About 25 cm (10 inches); the shortest | A C-shaped curve from the pyloric sphincter around the head of the pancreas. Mostly retroperitoneal. Bile and pancreatic juice enter through the major duodenal papilla on its inner wall |
| **Jejunum** | About 1 m (3 feet) | The middle region, mostly in the upper left abdomen, hanging on the mesentery. The name means empty, which is how it is found after death |
| **Ileum** | About 2 m (6 feet); the longest | The last region, mostly in the lower right abdomen, hanging on the mesentery. Ends at the ileocecal valve. Its wall holds Peyer patches |

- There is no sharp border between the jejunum and the ileum; one becomes the other along the mesentery.
- The ileum is the longest part, not the jejunum. Your slides give the duodenum as 12 inches; OpenStax and Tortora give about 25 cm (10 inches). Either is accepted.

### The wall of the small intestine

| Layer | In the small intestine |
|---|---|
| **Mucosa** | Simple columnar epithelium with several cell types: absorptive cells (enzymes on their surface, microvilli for absorption), goblet cells (mucus), and in the intestinal glands (crypts of Lieberkuhn) Paneth cells (lysozyme, a bactericidal enzyme, and phagocytosis) and enteroendocrine cells: S cells (secretin), CCK cells (cholecystokinin), and K cells (glucose-dependent insulinotropic peptide). Lamina propria of areolar connective tissue with MALT: solitary lymphatic nodules, most numerous in the distal ileum, and aggregated lymphatic follicles (Peyer patches) in the ileum. Muscularis mucosae of smooth muscle |
| **Submucosa** | In the duodenum, duodenal (Brunner) glands that secrete alkaline mucus to neutralize the acid in chyme. Lymphatic tissue sometimes extends down from the lamina propria |
| **Muscularis** | Two layers: a thicker inner circular layer and a thinner outer longitudinal layer |
| **Serosa** | Visceral peritoneum completely surrounds the small intestine, except the retroperitoneal part of the duodenum |

### Three levels of folding, largest to smallest

| Feature | What it is | Seen with |
|---|---|---|
| **Circular folds (plicae circulares)** | Permanent ridges of mucosa and submucosa running around the inside of the tube, from the proximal duodenum to the middle of the ileum. Some go all the way around, some part way. Unlike rugae they do not flatten when the tube fills. They increase the surface and make chyme spiral instead of running straight through | The naked eye |
| **Villi** | Finger-like projections of the mucosa, 20 to 40 per square millimeter, that give the lining a velvety look. Each has a core of lamina propria with an arteriole, a venule, a capillary network, and a lacteal | A light microscope |
| **Microvilli** | Tiny projections of the apical membrane of each absorptive cell, each a bundle of 20 to 30 actin filaments, about 200 million per square millimeter. Together they form the brush border, which also carries digestive enzymes | An electron microscope (a fuzzy edge on a light slide) |

- Fold, finger, fuzz. Circular folds are folds of the wall. Villi are fingers on the folds. Microvilli are the fuzz on each cell of each finger.
- The lacteal is the lymphatic capillary in the center of each villus. Most absorbed fats are too large to enter a blood capillary, so they enter the lacteal and ride the lymph before reaching the blood. Sugars and amino acids enter the blood capillaries.

### Two hormones from the small intestine

| Hormone | Released by | Trigger | Main effects |
|---|---|---|---|
| **Cholecystokinin (CCK)** | CCK cells of the intestinal glands | Chyme with amino acids and fatty acids | Pancreatic juice rich in enzymes; contraction of the gallbladder, squeezing bile down the cystic and common bile ducts; relaxation of the sphincter of Oddi; slower gastric emptying; a feeling of fullness (satiety) through the hypothalamus; growth of the pancreas; enhances secretin |
| **Secretin** | S cells of the intestinal glands | Acidic chyme | Pancreatic juice rich in bicarbonate, which buffers the acid; less gastric acid and gastric juice; growth of the pancreas; enhances CCK |

Together they keep the duodenum from being overloaded with more chyme than it can handle. Other gut hormones: motilin, substance P, and bombesin stimulate motility; vasoactive intestinal peptide (VIP) stimulates secretion of ions and water and inhibits gastric acid; gastrin-releasing peptide stimulates gastrin release; somatostatin inhibits it.

## Competency 117. Large intestine features

**You should be able to:** Identify the cecum, appendix, four parts of the colon, rectum, and anal canal along with the teniae coli, haustra, and omental appendices.

The large intestine is the last part of the canal, from the ileocecal valve to the anus, about 1.5 m (5 feet) long and about 6.5 cm (2.5 inches) across, the same in the living person and the cadaver. It frames the small intestine on three sides. It absorbs water, ions, and some vitamins, forms and stores feces, and empties them. Bacteria living in it break down leftover proteins and make some B vitamins and vitamin K. Haustral churning, peristalsis, and mass peristalsis drive the contents toward the rectum.

### The regions, in order

| Region | Where it is |
|---|---|
| **Cecum** | A blind pouch in the lower right abdomen hanging below the ileocecal valve. The first part of the large intestine |
| **Vermiform appendix** | A narrow, worm-shaped tube attached to the cecum, full of lymphatic tissue. Its own small mesentery, the mesoappendix, ties it to the ileum. Food never travels through it |
| **Ascending colon** | Up the right side to the liver. Retroperitoneal |
| **Hepatic (right colic) flexure** | The bend under the liver where the ascending colon turns left |
| **Transverse colon** | Across the upper abdomen from right to left, below the stomach. Hangs on the transverse mesocolon |
| **Splenic (left colic) flexure** | The bend near the spleen where the transverse colon turns down |
| **Descending colon** | Down the left side. Retroperitoneal |
| **Sigmoid colon** | The S-shaped part beginning near the left iliac crest, curving medially into the pelvis to join the rectum. Hangs on the sigmoid mesocolon |
| **Rectum** | The straight part in the pelvis, in front of the sacrum and coccyx, from the rectosigmoid junction to the rectoanal junction. Stores feces temporarily; its lower widened part, the ampulla, holds them until defecation. Three transverse rectal folds, one on the right and two on the left |
| **Anal canal** | The last 2 to 3 cm, passing through the pelvic floor to open at the anus. Its mucosa is in longitudinal folds, the anal columns, with arteries and veins. Lined with stratified squamous epithelium near the anus. Surrounded by the internal anal sphincter (smooth, involuntary) and the external anal sphincter (skeletal, voluntary), which keep the anus closed except during defecation |

- The colon has four parts: ascending, transverse, descending, sigmoid. The two flexures are bends between them, not separate parts.
- Trace it as a frame: up the right side, a bend at the liver, across the top, a bend at the spleen, down the left side, then an S into the pelvis.
- Rectum and anal canal lengths differ between sources, so you are asked where each one sits, not how long it is.

### The wall of the large intestine

| Layer | In the large intestine |
|---|---|
| **Mucosa** | Simple columnar epithelium with absorptive cells (water absorption) and many goblet cells (mucus to lubricate the passage), both in intestinal glands that run the full thickness of the mucosa. No circular folds and no villi, only microvilli on the absorptive cells, so far less absorption than the small intestine. Lamina propria with solitary lymph nodules |
| **Submucosa** | Areolar connective tissue |
| **Muscularis** | The outer longitudinal layer is gathered into three thick bands, the teniae coli, with little or no longitudinal muscle between them. The inner circular layer is a complete sheet |
| **Serosa** | Visceral peritoneum, with the omental appendices hanging from it |

### Three features found only on the large intestine

**Teniae coli**

Three flat ribbons of smooth muscle running the length of the cecum and colon: the outer longitudinal layer gathered into bands instead of a full sheet. They spread back into a complete layer at the rectum, so the rectum has no haustra. The three meet at the base of the appendix, which is one way to find it.

**Haustra**

The row of pouches that gives the colon its puckered look. The teniae are shorter than the wall and keep a tonic contraction, so the wall bunches up between them.

**Omental (epiploic) appendices**

Small fat-filled pouches of visceral peritoneum hanging from the surface of the colon, mostly along the teniae.

These three are how you tell large from small intestine at a glance on a model or the cadaver. Count the teniae: three, not two or four.

### Blood and nerves of the rectum

- Arteries: the superior rectal artery (from the inferior mesenteric artery) supplies the upper rectum; the right and left middle rectal arteries supply the lower rectum.
- Veins: a venous plexus in the wall drained by the rectal veins.
- Nerves: sympathetic fibers from the lumbar splanchnic nerves through the hypogastric plexus; parasympathetic and visceral sensory fibers through the pelvic splanchnic nerves.

## Competency 118. Trace the pathway of food

**You should be able to:** Trace a bite of food through every named segment of the alimentary canal in order from mouth to anus.

A bite of food moves through the alimentary canal in one direction only. Every segment below is one the food passes through, in order, with the sphincters at the borders where they sit.

1. Oral cavity: food comes in through the lips, is chewed by the teeth, mixed with saliva, and shaped into a bolus by the tongue.
2. Fauces: the tongue pushes the bolus back through the opening into the throat.
3. Oropharynx: the soft palate and uvula rise to seal the nasopharynx above.
4. Laryngopharynx: the epiglottis covers the larynx, so the bolus passes behind it toward the esophagus.
5. Upper esophageal sphincter relaxes.
6. Esophagus: waves of muscle contraction carry the bolus down through the thorax and the esophageal hiatus.
7. Lower esophageal sphincter relaxes.
8. Cardia of the stomach, then the body, where the food is churned with gastric juice into chyme.
9. Pyloric antrum, pyloric canal, pylorus: the funnel out of the stomach.
10. Pyloric sphincter lets small amounts of chyme through.
11. Duodenum: bile and pancreatic juice enter at the major duodenal papilla.
12. Jejunum: digestion continues and most nutrients are absorbed.
13. Ileum: absorption continues.
14. Ileocecal sphincter (valve) lets the remaining material into the large intestine.
15. Cecum, then ascending colon, hepatic flexure, transverse colon, splenic flexure, descending colon, sigmoid colon.
16. Rectum: feces are stored.
17. Anal canal, through the internal and external anal sphincters, to the anus.

- Three named structures are off the path. The nasopharynx carries only air and is sealed during a swallow. The fundus bulges above the cardia, so food can collect there without passing through it. The appendix is a blind tube off the cecum.
- The cecum comes right after the ileum. Do not jump from the ileum to the ascending colon.
- Group the path as the six organs first (mouth, pharynx, esophagus, stomach, small intestine, large intestine), then fill in the named parts inside each one and put a sphincter at each border.
- Most absorption happens in the small intestine. Water is absorbed and feces form in the large intestine.

## Competency 119. Clinical anatomy of canal disorders

**You should be able to:** Match GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, IBD, hemorrhoids, and colorectal cancer to the canal structure each one affects.

Each disorder starts in a specific structure. If you know where that structure sits and what layers it has, you can predict where the patient feels it and how serious it can become.

| Disorder | Structure affected | What changes |
|---|---|---|
| **Gastroesophageal reflux disease (GERD)** | Lower esophageal sphincter and lower esophagus | The sphincter does not close well, so acidic stomach contents flow back up. The stratified squamous lining of the esophagus has no protection against acid and becomes inflamed. Felt as heartburn, a burning behind the sternum, because the esophagus lies in the thorax just behind it |
| **Hiatal hernia** | Esophageal hiatus and upper stomach | Part of the stomach pushes up through the hiatus into the thorax. The diaphragm loses its grip on the lower esophageal sphincter, so reflux often follows |
| **Peptic ulcer** | Mucosa of the stomach or duodenum | An open sore where the mucosa has worn away. Common causes: infection with Helicobacter pylori and regular use of NSAID pain relievers. It deepens layer by layer: through the submucosa, where it can open a vessel and bleed, through the muscularis, and through the serosa, where it perforates and the contents spill into the peritoneal cavity |
| **Appendicitis** | Vermiform appendix | The lumen blocks, the appendix swells and inflames, and pain lands in the lower right abdomen where the cecum sits. If it bursts, its contents spill into the peritoneal cavity: peritonitis |
| **Diverticulosis** | Wall of the colon, most often the sigmoid | Small pouches (diverticula) of mucosa and submucosa bulge out through weak spots in the muscularis. When the pouches inflame, it is diverticulitis |
| **Inflammatory bowel disease (IBD)** | Wall of the small and large intestine | Long-term inflammation of the wall. Crohn disease can affect any part from mouth to anus, most often the ileum and colon, through every layer. Ulcerative colitis affects only the colon and rectum, mostly the mucosa |
| **Hemorrhoids** | Veins in the wall of the anal canal and lower rectum | The veins swell and stretch into soft bulges. External hemorrhoids are thrombosed veins of the external plexus, often with pregnancy or constipation, and painful because they have somatic innervation. Internal hemorrhoids are dilated veins of the internal rectal plexus above the dentate line, visceral innervation, so painless unless they ulcerate |
| **Colorectal cancer** | Mucosa of the colon or rectum | Lining cells grow out of control, usually starting as a polyp, then growing through the deeper layers of the wall |

### What this means for patients

- Where a problem sits predicts the symptom. A weak lower esophageal sphincter burns behind the sternum. An inflamed appendix hurts in the lower right.
- The four-layer wall explains how a problem gets worse. Ulcers, diverticula, IBD, and cancer all become more serious as they reach deeper layers, and a perforation through the serosa turns a gut problem into peritonitis.
- GERD and hiatal hernia are two different things that often travel together: one is a weak sphincter, the other is stomach through the diaphragm.

# Blood flow of the abdomen and lower limb

The written notes for the blood flow lecture: the abdominal aorta and its branches, the inferior vena cava and its tributaries, the hepatic portal system, and the vessels and nerves of the lower limb. Module 4 competencies 126 and 127, with nine worked questions.

## Competency 126. The abdominal aorta and its branches

**You should be able to:** Identify the major branches of the abdominal aorta and the vessels of the hepatic portal system on the cadaver.

The abdominal aorta is the aorta below the diaphragm. It enters the abdomen through the aortic hiatus at about T12, runs down the front of the lumbar vertebrae just left of the midline with the inferior vena cava on its right, and ends at about L4 by dividing into the right and left common iliac arteries. Its branches come in two groups: unpaired branches leave the front and feed the digestive organs; paired branches leave the sides and feed the kidneys, the gonads, and other structures that come in twos.

| Branch | About where it leaves | What it supplies |
|---|---|---|
| **Celiac trunk (unpaired)** | T12, just below the diaphragm | Short; splits almost at once into the left gastric, splenic, and common hepatic arteries |
| **Left gastric artery** | From the celiac trunk | The stomach along the lesser curvature and the lower esophagus |
| **Splenic artery** | From the celiac trunk | Runs left along the top of the pancreas to the spleen, with branches to the pancreas and stomach |
| **Common hepatic artery** | From the celiac trunk | Runs right; gives the hepatic artery proper to the liver and gallbladder, plus branches to the stomach, duodenum, and pancreas |
| **Superior mesenteric artery (unpaired)** | L1 | Runs in the mesentery to most of the small intestine, part of the pancreas, the cecum, appendix, ascending colon, and most of the transverse colon |
| **Renal arteries (paired)** | L1 to L2 | The kidneys |
| **Gonadal arteries (paired)** | L2 | The testes (testicular arteries) or the ovaries (ovarian arteries) |
| **Inferior mesenteric artery (unpaired)** | L3 | The end of the transverse colon, the descending colon, the sigmoid colon, and the upper rectum |
| **Right and left common iliac arteries** | L4, where the aorta ends | The pelvis and lower limbs, through the internal and external iliac arteries |

- The three gut branches go in the same order as the gut. Celiac feeds the top (stomach, liver, spleen, pancreas, start of the duodenum). Superior mesenteric feeds the middle (small intestine and the colon through most of the transverse). Inferior mesenteric feeds the end (the rest of the colon and the upper rectum).
- The three unpaired branches are the celiac trunk, superior mesenteric, and inferior mesenteric. Renal and gonadal arteries are paired.
- The hepatic artery proper rides up in the free edge of the lesser omentum to the porta hepatis.

### Arteries of the pelvis

- Each common iliac artery divides in front of the sacroiliac joint. The internal iliac artery dives into the pelvis and supplies the bladder, rectum, reproductive organs, pelvic walls, gluteal muscles, and external genitalia. The external iliac artery runs along the psoas major, passes under the inguinal ligament, and becomes the femoral artery.
- The medial umbilical ligament is a fibrous cord on the inside of the anterior abdominal wall running from the pelvis toward the umbilicus. It is what remains of the fetal umbilical artery, a branch of the internal iliac. The part nearest the internal iliac stays open and supplies the bladder.

## Competency 126. The inferior vena cava and the veins of the abdomen

The inferior vena cava (IVC) forms at about L5 where the right and left common iliac veins join. It runs up the right side of the aorta, passes through the central tendon of the diaphragm at about T8, and empties into the right atrium.

| Vein | Drains | Empties into |
|---|---|---|
| **External iliac vein** | The lower limb; it is the femoral vein above the inguinal ligament | Common iliac vein |
| **Internal iliac vein** | The pelvis | Common iliac vein |
| **Right and left common iliac veins** | The pelvis and lower limb | Join to form the inferior vena cava |
| **Renal veins** | The kidneys. The left renal vein is longer because it crosses in front of the aorta to reach the IVC | Inferior vena cava |
| **Right gonadal vein** | The right testis or ovary | Inferior vena cava |
| **Left gonadal vein** | The left testis or ovary | Left renal vein |
| **Hepatic veins** | The liver, after the sinusoids and central veins | Inferior vena cava, just below the diaphragm |
| **Inferior vena cava** | Most of the body below the diaphragm | Right atrium |

- The gonadal veins are not symmetric. Right into the IVC, left into the left renal vein.
- The stomach, intestines, spleen, and pancreas are not on this list. Their blood goes to the liver first, through the hepatic portal system.

## Competency 126. The hepatic portal system

A portal system is a route in which blood passes through two capillary beds in a row before it returns to the heart. In the hepatic portal system the first bed is in the walls of the stomach, intestines, spleen, and pancreas, and the second bed is the sinusoids of the liver. The hepatic portal vein connects them.

Everything absorbed from the gut, nutrients and whatever came with them, is processed by the liver before it reaches the rest of the body. It is also why cancers of the digestive tract so often spread to the liver first.

| Vessel | Drains | Empties into |
|---|---|---|
| **Superior mesenteric vein** | The small intestine, cecum, ascending colon, and most of the transverse colon; runs beside the superior mesenteric artery | Joins the splenic vein to form the hepatic portal vein |
| **Inferior mesenteric vein** | The descending colon, sigmoid colon, and upper rectum | Usually the splenic vein |
| **Splenic vein** | The spleen, pancreas, and part of the stomach; runs along the back of the pancreas | Joins the superior mesenteric vein to form the hepatic portal vein |
| **Hepatic portal vein** | All of the above. Forms behind the neck of the pancreas, runs up in the free edge of the lesser omentum, and enters the liver at the porta hepatis | The hepatic sinusoids |
| **Hepatic veins** | The liver, from the central veins | Inferior vena cava |

### A glucose molecule, from a villus to the heart

1. Capillaries of an intestinal villus in the jejunum. The glucose enters the blood here: the first capillary bed.
2. Small veins of the intestinal wall and mesentery.
3. Superior mesenteric vein.
4. Hepatic portal vein, formed where the superior mesenteric and splenic veins join.
5. Porta hepatis, then a portal vein branch in a portal triad.
6. Hepatic sinusoids, past the hepatocytes: the second capillary bed.
7. Central vein of the lobule.
8. Hepatic veins, out of the back of the liver.
9. Inferior vena cava, through the diaphragm at T8.
10. Right atrium.

- The hepatic portal vein carries blood into the liver. The hepatic veins carry it out. Portal in, hepatic out.
- Two things enter the porta hepatis (the portal vein and the hepatic artery proper) and the hepatic ducts leave it. The hepatic veins do not use the porta hepatis; they leave the posterior surface straight into the vena cava.
- Portal vein blood is nutrient-rich and oxygen-poor. The hepatic artery brings the oxygen. They mix in the sinusoids and flow in toward the central vein, while bile flows out toward the triads.

## Competency 127. Arteries and veins of the lower limb

**You should be able to:** Identify the femoral and lower-limb blood vessels and the major nerves of the abdomen and lower limb on the cadaver.

In the lower limb, arteries, veins, and nerves run together in bundles between the muscles you already know. Use the muscles as landmarks. When you name a vessel, say right or left and artery or vein.

### One artery, renamed at each landmark

1. External iliac artery: along the pelvic brim from the common iliac.
2. Femoral artery: the same vessel after it passes under the inguinal ligament. Its pulse is felt about halfway between the anterior superior iliac spine and the pubic symphysis. It runs through the femoral triangle, gives off the deep femoral artery to most of the thigh muscles, then runs down under the sartorius.
3. Popliteal artery: the same vessel after it passes through the adductor hiatus in adductor magnus to reach the back of the knee.
4. Anterior and posterior tibial arteries: the popliteal divides just below the knee. The anterior tibial goes forward into the front of the leg and becomes the dorsalis pedis artery on the top of the foot. The posterior tibial, the larger branch, runs down the back of the leg deep to the gastrocnemius and soleus beside the tibial nerve, gives off the fibular artery to the lateral leg, passes behind the medial malleolus (second pulse point), and divides into the plantar arteries in the sole.

**Femoral triangle**

The space just below the inguinal ligament, bounded by the inguinal ligament above, the sartorius laterally, and the adductor longus medially. From lateral to medial: femoral nerve, femoral artery, femoral vein, empty space, lymphatics (NAVEL). The vein is medial to the artery.

**Adductor hiatus**

The opening in adductor magnus through which the femoral artery and vein pass to the back of the knee, where their names change to popliteal.

### Veins of the lower limb

| Vessel | Deep or superficial | Course and landmarks |
|---|---|---|
| **Posterior and anterior tibial veins, popliteal vein, femoral vein** | Deep | Run beside the arteries and share their names. The tibial veins join into the popliteal vein, which becomes the femoral vein up the thigh (medial to the artery in the triangle), which becomes the external iliac vein above the inguinal ligament. The deep veins carry most of the blood out of the limb |
| **Great saphenous vein** | Superficial | The longest vein in the body. Starts on the medial side of the top of the foot, passes in front of the medial malleolus, runs up the medial leg and thigh in the fat under the skin, and empties into the femoral vein just below the inguinal ligament. Surgeons use it as a graft in coronary bypass surgery |
| **Small saphenous vein** | Superficial | Up the back of the calf into the popliteal vein |

- The name changes at three landmarks on the way to the foot: the diaphragm (thoracic to abdominal aorta), the inguinal ligament (external iliac to femoral), and the adductor hiatus (femoral to popliteal).
- The great saphenous vein is superficial and medial. The femoral vein is the deep one.

## Competency 127. Nerves of the abdomen and lower limb

Only the femoral nerve is on your Module 4 structure list, and the plexus notes also name the obturator nerve. The rest are here so you can recognize what you see on the posterior abdominal wall and in the limb. You identify nerves by location; you are not asked which nerve supplies each muscle.

### On the posterior abdominal wall

The lumbar plexus forms inside the psoas major from the anterior rami of L1 to L4. Its branches come out of the psoas and cross the posterior wall, which is where you see them on the cadaver.

| Nerve | Spinal nerves | Where you see it | What it supplies |
|---|---|---|---|
| **Iliohypogastric** | L1 | Crossing the posterior wall toward the iliac crest | Lower abdominal wall muscles, skin above the pubis |
| **Ilioinguinal** | L1 | Just below the iliohypogastric, then through the inguinal canal | Lower abdominal wall, skin of the upper medial thigh and external genitalia |
| **Genitofemoral** | L1 to L2 | Down the front of the psoas major | Skin of the external genitalia and a patch of upper anterior thigh |
| **Lateral femoral cutaneous** | L2 to L3 | Across the iliacus toward the anterior superior iliac spine | Skin of the lateral thigh |
| **Femoral** | L2 to L4 | Out of the lateral side of the psoas, in the groove between psoas major and iliacus | Anterior thigh muscles and skin |
| **Obturator** | L2 to L4 | Out of the medial side of the psoas, to the obturator foramen | Adductor muscles and skin of the medial thigh |

- Most of the anterolateral abdominal wall is supplied by the lower intercostal nerves (T7 to T11) and the subcostal nerve (T12); the lumbar plexus supplies only the lowest strip above the pubis and the groin.
- The anterior and posterior vagal trunks (vagus, cranial nerve X) enter the abdomen on the esophagus through the esophageal hiatus and supply the stomach and other abdominal organs.

### In the lower limb

| Nerve | Plexus | Course | Muscles supplied |
|---|---|---|---|
| **Femoral nerve** | Lumbar plexus (L2 to L4) | Under the inguinal ligament, lateral to the femoral artery, into the femoral triangle | Anterior thigh: quadriceps femoris and sartorius; a branch continues to the skin of the medial leg |
| **Sciatic nerve** | Sacral plexus (L4 to S3) | Out the greater sciatic foramen, deep to gluteus maximus, down the back of the thigh deep to the hamstrings; the largest nerve in the body | The hamstrings; splits above the knee into the tibial and common fibular nerves |
| **Tibial nerve** | Sacral, from the sciatic | Straight down through the back of the knee and the posterior leg deep to gastrocnemius and soleus, beside the posterior tibial artery, behind the medial malleolus into the sole | Posterior leg muscles and the sole of the foot |
| **Common fibular nerve** | Sacral, from the sciatic | Along the biceps femoris tendon to the lateral knee, around the neck of the fibula just under the skin, where it splits into superficial and deep branches | Superficial branch: lateral leg (fibularis longus and brevis). Deep branch: anterior leg (tibialis anterior, extensor hallucis longus, extensor digitorum longus, fibularis tertius) |

- The tibial and common fibular nerves come off the sciatic nerve, not straight off the plexus.
- Because the common fibular nerve lies against the neck of the fibula just under the skin, a blow or a tight cast there can injure it and weaken the muscles that lift the foot: foot drop.

# Endocrine anatomy

The written notes for the endocrine lecture: the major glands and where each sits, the pituitary and its two lobes, the thyroid and parathyroid glands, the adrenal zones, and the pancreas, gonads, and other organs with endocrine tissue. Module 4 competencies 128 to 132, with nine worked questions.

## Before the competencies: what an endocrine gland is

The endocrine system is a scattered set of glands and tissues that release hormones, chemical messengers that travel in the blood. Unlike most organ systems, most of its glands are not physically connected to each other. They sit in the head, neck, thorax, abdomen, pelvis, and scrotum, and the blood is what links them.

An endocrine gland is a ductless gland. Its cells release their product into the surrounding interstitial fluid, and from there it enters the blood. That is why endocrine glands have such a rich supply of capillaries. An exocrine gland is different: it has a duct that carries its product onto a surface or into a cavity, the way the salivary glands empty into the mouth. Some organs, such as the pancreas, have both kinds of tissue.

**Endocrine system**
The glands and tissues that release hormones into the blood to signal distant cells.

**Hormone**
A chemical messenger released by an endocrine cell that travels in the blood to target cells.

**Endocrine gland**
A ductless gland that releases its product into the surrounding fluid and the blood.

**Exocrine gland**
A gland that releases its product through a duct onto a surface or into a cavity. Not part of the endocrine system.

**Target cell**
A cell that carries the receptor for a given hormone and so responds to it.

- Ductless and into the blood means endocrine. Through a duct and onto a surface means exocrine.
- Because hormones travel in the blood, an endocrine gland can be far from the cells it controls.

## Competency 128. Major endocrine glands and locations

**You should be able to:** Name the major endocrine glands and locate each one in the body from the hypothalamus to the gonads.

Learn the major glands from the top of the body down. Three are in the brain or hang from it: the hypothalamus, the pituitary gland, and the pineal gland. Two are in the neck: the thyroid gland and the parathyroid glands behind it. The thymus is in the thorax. The adrenal glands and the pancreas are in the abdomen, and the endocrine tissue of the pancreas is the pancreatic islets. The gonads are the ovaries in the female pelvis and the testes in the male scrotum.

### Where each gland sits

| Gland | Body region | Location |
|---|---|---|
| **Hypothalamus** | Brain | The lowest part of the diencephalon, below the thalamus. It forms the floor and lower walls of the third ventricle. It links the nervous and endocrine systems and controls the pituitary |
| **Pituitary gland** | Brain | A pea-sized gland in the sella turcica of the sphenoid bone, hanging below the hypothalamus by the infundibulum |
| **Pineal gland** | Brain | A small, cone-shaped gland in the epithalamus, projecting from the posterior roof of the third ventricle |
| **Thyroid gland** | Neck | A butterfly-shaped gland in the anterior neck just below the larynx. Its two lobes lie beside the trachea and its isthmus crosses in front of it. The largest purely endocrine gland |
| **Parathyroid glands** | Neck | Usually four tiny glands on the posterior surface of the thyroid lobes, two on each side |
| **Thymus** | Thorax | In the mediastinum, behind the sternum and above the heart. Large in childhood, shrinks after puberty |
| **Adrenal glands** | Abdomen | Also called the suprarenal glands. One on the superior pole of each kidney, on the posterior abdominal wall behind the peritoneum |
| **Pancreatic islets** | Abdomen | Small clusters of endocrine cells scattered through the pancreas, which lies behind the stomach |
| **Ovaries** | Pelvis | In the pelvic cavity, one on each side of the uterus |
| **Testes** | Scrotum | In the scrotum, outside the abdominopelvic cavity |

- Memory tip: learn the glands from top to bottom by body region. Brain: hypothalamus, pituitary, pineal. Neck: thyroid and parathyroids. Chest: thymus. Abdomen: adrenals and pancreas. Pelvis: ovaries, or testes in the scrotum.
- The pineal gland sits at the back of the third ventricle and the pituitary hangs below the hypothalamus, in front and below. They are not in the same spot.
- Four of the glands come in pairs or more: the parathyroids (usually four, not two), the adrenals, the ovaries, and the testes. The pancreatic islets are many small clusters inside one pancreas.
- The thyroid is the largest gland in the body that is purely endocrine. The thymus is large in childhood and shrinks after puberty.

### What this means for patients

- The pituitary gland sits in the bony sella turcica, so a growing pituitary tumor usually expands upward out of the sella. Directly above it is the optic chiasm, and pressure there can cause a loss of peripheral vision. That vision change can be the first sign that a gland the size of a pea has enlarged.
- The thyroid gland lies just under the skin of the anterior neck, so an enlarged thyroid, a goiter, is visible and can be felt. The same superficial position lets a clinician palpate the gland and makes thyroid surgery accessible, though the tiny parathyroid glands on its back surface must be carefully spared.
- Each adrenal gland rests on the superior pole of a kidney. Imaging ordered for the kidney also captures the adrenal, which is why an adrenal mass is sometimes found by accident.

## Competency 129. Pituitary gross anatomy and lobes

**You should be able to:** Describe the pituitary in the sella turcica, its infundibulum link to the hypothalamus, and compare the anterior and posterior lobes by tissue type.

The pituitary gland, also called the hypophysis, is about the size of a pea. It sits in the sella turcica, a saddle-shaped pocket in the body of the sphenoid bone in the middle of the base of the skull. The bone forms the floor and the front and back walls of this pocket and protects the gland. The optic chiasm lies just above and in front of it.

The pituitary hangs from the hypothalamus by a slender stalk called the infundibulum. The infundibulum is the physical link between the brain and the gland. Nerve fibers from the hypothalamus run down through it, and small blood vessels run down along it.

**Hypophysis**
Another name for the pituitary gland.

**Sella turcica**
The saddle-shaped pocket of the sphenoid bone that cradles and protects the pituitary.

**Infundibulum**
The slender stalk that connects the pituitary to the hypothalamus above it.

**Hypothalamo-hypophyseal tract**
The bundle of nerve fibers that runs from the hypothalamus down through the infundibulum to the posterior lobe.

**Hypophyseal portal system**
The blood vessels that carry hypothalamic signals directly to the anterior lobe.

### The two lobes

The pituitary has two lobes that develop from two different tissues in the embryo, so it is really two glands joined together. The anterior lobe, or adenohypophysis, grows up from the roof of the embryonic mouth. It is made of glandular epithelial tissue. Its cells are arranged in cords and clusters with wide capillaries running between them. The anterior lobe is the larger of the two.

The posterior lobe, or neurohypophysis, grows down from the floor of the brain. It is made of nervous tissue and is an extension of the hypothalamus itself. It contains the axons of neurons whose cell bodies sit in the hypothalamus, along with supporting glial cells called pituicytes. The posterior lobe makes no hormones of its own. It stores and releases what the hypothalamic neurons make and send down their axons.

Because the tissues differ, each lobe connects to the hypothalamus in a different way. The posterior lobe connects by nerve fibers, the hypothalamo-hypophyseal tract, which runs down through the infundibulum. The anterior lobe has no nerve link to the hypothalamus. It connects through blood vessels, the hypophyseal portal system, which has three parts in order:

1. Primary capillary plexus: a capillary bed at the base of the hypothalamus, where the infundibulum begins, picks up hypothalamic signals.
2. Hypophyseal portal veins: small veins carry the blood down along the infundibulum.
3. Secondary capillary plexus: a second capillary bed in the anterior lobe delivers the signals to the glandular cells.

| Feature | Anterior lobe | Posterior lobe |
|---|---|---|
| **Other name** | Adenohypophysis | Neurohypophysis |
| **Tissue type** | Glandular epithelial tissue | Nervous tissue, an extension of the hypothalamus |
| **Develops from** | An upgrowth of the roof of the embryonic mouth | A downgrowth of the floor of the brain |
| **Link to the hypothalamus** | Blood vessels, the hypophyseal portal system | Nerve fibers, the hypothalamo-hypophyseal tract, through the infundibulum |
| **On a slide** | Densely packed gland cells, many stained pink or blue, in cords with capillaries between them | Pale, fibrous-looking tissue of axons and scattered pituicyte nuclei |

- Memory tip: "neuro" means nervous tissue and "adeno" means gland. The posterior lobe is nervous tissue, so it connects by nerve fibers. The anterior lobe is glandular epithelium, so it connects by blood vessels.
- The tissue type predicts the link. Nervous tissue links by nerve fibers; glandular tissue links by blood vessels. The nerve fibers go to the posterior lobe and the blood vessels go to the anterior lobe, never the other way around.
- A pituitary tumor grows up and out of the sella turcica, because bone closes the floor and walls, and the first thing it presses on is the optic chiasm.

## Competency 130. Thyroid and parathyroid anatomy

**You should be able to:** Identify the thyroid lobes, isthmus, follicles, colloid, follicular and parafollicular cells, and the parathyroid chief and oxyphil cells.

### The thyroid gland, gross anatomy

The thyroid gland is a butterfly-shaped gland in the anterior neck, just below the thyroid cartilage of the larynx. It has a right lobe and a left lobe. Each lobe lies against the side of the larynx and trachea. The two lobes are joined across the midline by a narrow bridge of thyroid tissue called the isthmus, which crosses in front of the upper rings of the trachea, just below the cricoid cartilage. A thin connective tissue capsule covers the gland. The thyroid is the largest gland in the body that is purely endocrine, and it has a very rich blood supply.

### The thyroid gland, on the slide

Under the microscope, the thyroid is made of thousands of tiny hollow spheres called thyroid follicles. Each follicle is a ring of cells around a center filled with colloid. Colloid is a thick, protein-rich fluid. It stains pink and fills the center of each follicle.

The wall of each follicle is a single layer of follicular cells. They form a simple cuboidal epithelium that sits around the colloid. Follicular cells are the main hormone-making cells of the thyroid. Between the follicles, or tucked into the outer edge of a follicle wall, are the parafollicular cells, also called C cells. They are larger and paler than follicular cells, and they do not touch the colloid. They occur alone or in small groups.

| Structure | What it is | How to find it |
|---|---|---|
| **Right and left lobes** | The two lateral parts of the gland | On each side of the larynx and trachea, below the thyroid cartilage |
| **Isthmus** | The bridge of thyroid tissue that connects the two lobes | Crossing the front of the upper trachea in the midline |
| **Thyroid follicles** | Microscopic hollow spheres that make up the gland | Rings of cells packed together across the slide |
| **Colloid** | The protein-rich fluid stored in the center of each follicle | Pink material filling each ring |
| **Follicular cells** | The simple cuboidal cells forming the wall of a follicle | The single layer of cells lining each ring of colloid |
| **Parafollicular (C) cells** | Endocrine cells between the follicles | Larger, pale cells between follicles or at the outer edge of a follicle wall, not touching the colloid |

- The lobes and isthmus are things you identify on the model or the cadaver. Follicles, colloid, and C cells are too small to see by eye, so those are slide structures only.
- Colloid fills the inside of the follicle. The C cells sit between follicles or at the outer edge of the wall, and they never touch the colloid.

### The parathyroid glands

The parathyroid glands are small, round glands, each about the size of a grain of rice. Most people have four, a superior and an inferior gland on the posterior surface of each thyroid lobe. Each one has its own thin capsule that separates it from the thyroid tissue. The number and exact position vary from person to person.

Parathyroid tissue has no follicles and no colloid. Its cells are packed in cords and clusters with capillaries between them. Two cell types are seen on a slide. Chief cells (also called principal cells) are the small, numerous cells that make up most of the gland. They are the hormone-making cells of the parathyroid. Oxyphil cells are larger and much fewer. Their cytoplasm is packed with mitochondria, so it stains dark pink. Oxyphil cells become more common with age. Their function is not fully understood.

| Structure | What it is | How to find it |
|---|---|---|
| **Parathyroid glands** | Usually four small, round glands on the posterior surface of the thyroid lobes | Two on the back of each lobe, each in its own capsule |
| **Chief cells** | The main hormone-making cells of the parathyroid glands | Small, numerous cells with round nuclei and pale cytoplasm, making up most of the tissue |
| **Oxyphil cells** | Larger, fewer cells whose function is not fully understood | Larger cells with dark pink, granular cytoplasm, alone or in small groups among the chief cells |

- Memory tip: follicles and colloid mean thyroid. Packed cords of cells with no colloid mean parathyroid. On a parathyroid slide, the small, plentiful cells are chief cells and the big pink ones are oxyphil cells.

### What this means for patients

- The parathyroid glands are tiny and embedded on the back of the thyroid, so during thyroid surgery they can be damaged or removed along with it. A surgeon has to find and spare them.
- Because the thyroid sits just under the skin of the front of the neck, an enlarged thyroid (a goiter) can be seen and felt.

## Competency 131. Adrenal gland zones

**You should be able to:** Identify the adrenal cortex zones in order and the adrenal medulla with its chromaffin cells.

An adrenal gland, also called a suprarenal gland, sits on the superior pole of each kidney. Both adrenal glands lie on the posterior abdominal wall behind the peritoneum, enclosed with the kidney in fat and the renal fascia. The right gland is shaped like a pyramid. The left gland is shaped like a crescent. Each gland is covered by a connective tissue capsule.

Each adrenal gland is two glands in one. The outer adrenal cortex makes up most of the gland. It is glandular epithelial tissue that develops from mesoderm. The inner adrenal medulla is a small core in the center. It is modified nervous tissue that develops from the neural crest, the same embryonic tissue that forms the sympathetic ganglia. On a slide the cortex looks like layered bands and the medulla looks like a different tissue in the middle.

**Capsule**
The connective tissue covering of the gland.

**Adrenal cortex**
The outer region and the bulk of the gland. Glandular epithelial tissue arranged in three zones.

**Adrenal medulla**
The inner core. Modified nervous tissue that belongs to the sympathetic nervous system.

### The three zones of the cortex, from the capsule inward

The cortex is built in three zones. Each zone is named for how its cells are arranged. Follow them in order, from the capsule inward to the medulla.

1. Zona glomerulosa: the thin outer zone, just under the capsule. Its cells form small rounded clusters and arches. "Glomerulus" means a little ball.
2. Zona fasciculata: the thick middle zone, the widest of the three. Its cells line up in long, straight columns running toward the center. "Fascicle" means a bundle. The cells look pale and foamy because they hold many lipid droplets.
3. Zona reticularis: the thin inner zone, against the medulla. Its cells form a branching network. "Reticulum" means a network. These cells stain darker than the fasciculata.

### The adrenal medulla and chromaffin cells

The medulla is made of chromaffin cells. These are large, round to many-sided cells arranged in clusters around blood vessels. Chromaffin cells are modified sympathetic neurons. They have no axons or dendrites, and they release their product straight into the blood. They are supplied by preganglionic sympathetic nerve fibers, so the medulla works like a sympathetic ganglion. Their name comes from the brown color they turn when stained with chromium salts.

| Layer | Part of | Position and thickness | Tissue type | How the cells are arranged |
|---|---|---|---|---|
| **Capsule** | Covering | Outermost, thin | Dense connective tissue | A connective tissue sheet |
| **Zona glomerulosa** | Cortex | Outer, thin | Glandular epithelial tissue | Rounded clusters and arches |
| **Zona fasciculata** | Cortex | Middle, the thickest of the three | Glandular epithelial tissue | Long, straight columns of pale, foamy cells |
| **Zona reticularis** | Cortex | Inner, against the medulla, thinner than the fasciculata | Glandular epithelial tissue | A branching network of darker cells |
| **Adrenal medulla** | Medulla | The core | Modified nervous tissue | Clusters of large chromaffin cells around blood vessels |

- Memory tip: GFR, from the outside in. Glomerulosa, Fasciculata, Reticularis. Then the medulla, full of chromaffin cells, in the center.
- The middle zone is the thickest. If your drawing has the glomerulosa as the widest band, the order or the widths are wrong.
- The three cortex zones are all glandular tissue. The medulla is the only part that is nervous tissue.

### What this means for patients

- The adrenal gland rests on the top of the kidney, so imaging ordered for the kidney usually shows the adrenal as well. That is how an adrenal mass is sometimes found by accident.

## Competency 132. Pancreas, gonads, and organs with endocrine tissue

**You should be able to:** Locate the pancreatic islets and gonads and name other organs such as the heart, kidneys, and skin that contain endocrine tissue.

### The pancreas and its islets

The pancreas is a long, soft gland in the upper abdomen. It lies behind the stomach, against the posterior abdominal wall and mostly behind the peritoneum. Its head sits in the C-shaped curve of the duodenum. Its body crosses behind the stomach, and its tail reaches toward the spleen on the left.

The pancreas is a mixed gland, with both exocrine and endocrine tissue. Most of it is exocrine tissue. This tissue is made of acini, clusters of cells that make digestive enzymes and send them through ducts to the duodenum. Scattered among the acini are about 1 to 2 million small islands of endocrine cells called the pancreatic islets, or islets of Langerhans. The islets have no ducts. They release their hormones into the many capillaries that run through them. On a slide, an islet is a round, pale-staining patch of cells surrounded by darker-staining acini.

### The gonads

The gonads are mixed organs too. They make gametes, and they also contain endocrine cells. The ovaries are the paired female gonads. They sit in the pelvic cavity, one on each side of the uterus, against the lateral pelvic wall. Their endocrine tissue is the cells of the developing ovarian follicles and the corpus luteum that forms after ovulation. The testes are the paired male gonads. They sit in the scrotum, outside the abdominopelvic cavity. Their endocrine tissue is the interstitial cells (Leydig cells), which lie in the spaces between the seminiferous tubules.

| Structure | Location | Endocrine part |
|---|---|---|
| **Pancreas** | Upper abdomen behind the stomach; head in the curve of the duodenum, tail toward the spleen | Both exocrine and endocrine; a mixed gland |
| **Pancreatic islets** | Scattered throughout the pancreas | The endocrine cell clusters (islets of Langerhans) |
| **Acini** | Most of the pancreas | None. The exocrine clusters that make digestive enzymes |
| **Ovaries** | Pelvic cavity, one on each side of the uterus | Cells of the ovarian follicles and the corpus luteum |
| **Testes** | Scrotum | Interstitial (Leydig) cells between the seminiferous tubules |

- The islets are the endocrine part of the pancreas and the acini are the exocrine part. Islets are pale, acini are dark, and the islets are the small patches.
- These notes call three organs mixed organs: the pancreas, the ovaries, and the testes. An organ is mixed when it does an endocrine job alongside another role, such as making digestive enzymes or making gametes.

### Other organs with endocrine tissue

Many organs whose main job is something else still contain endocrine cells. These organs are not counted as endocrine glands, and they are not called mixed organs either, but their endocrine cells still release hormones into the blood. The heart has hormone-making cells in the muscle of the atrial walls. The kidneys have endocrine cells in the renal cortex. The skin takes part too: cells of the epidermis begin making a hormone when the skin is exposed to sunlight. The stomach and small intestine, the liver, adipose tissue, and the placenta during pregnancy also contain endocrine cells.

| Organ | Where it sits | Where the endocrine cells are |
|---|---|---|
| **Hypothalamus** | Brain, floor and walls of the third ventricle | Neurosecretory cells. A neuroendocrine organ that controls the pituitary gland |
| **Heart** | Chest, in the mediastinum | Cardiac muscle cells in the walls of the atria |
| **Kidneys** | Abdomen, against the posterior wall | Endocrine cells in the renal cortex, including the juxtaglomerular cells in the walls of the afferent arterioles |
| **Skin** | Body surface | Keratinocytes of the epidermis, which begin making a hormone when exposed to sunlight |
| **Stomach and small intestine** | Abdomen | Enteroendocrine cells scattered in the mucosa |
| **Liver** | Upper right abdomen | Hepatocytes, the main liver cells |
| **Adipose tissue** | Under the skin and around organs | Adipocytes (fat cells) |
| **Placenta** | In the uterus during pregnancy | Placental cells. A temporary organ, because it exists only during pregnancy |

- The heart, kidneys, and skin contain endocrine tissue, but they are not endocrine glands and these notes do not count them as mixed organs. Only the pancreas, ovaries, and testes get that label.
- The placenta is the one temporary organ on the list. It forms during pregnancy and is gone afterward.